MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIPLEXER, DIGITAL, KEY CODE AHD

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic multiplexer module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787	-	Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1665	-	Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5963

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* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

* 3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.118 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance and requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be AHD.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/78-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

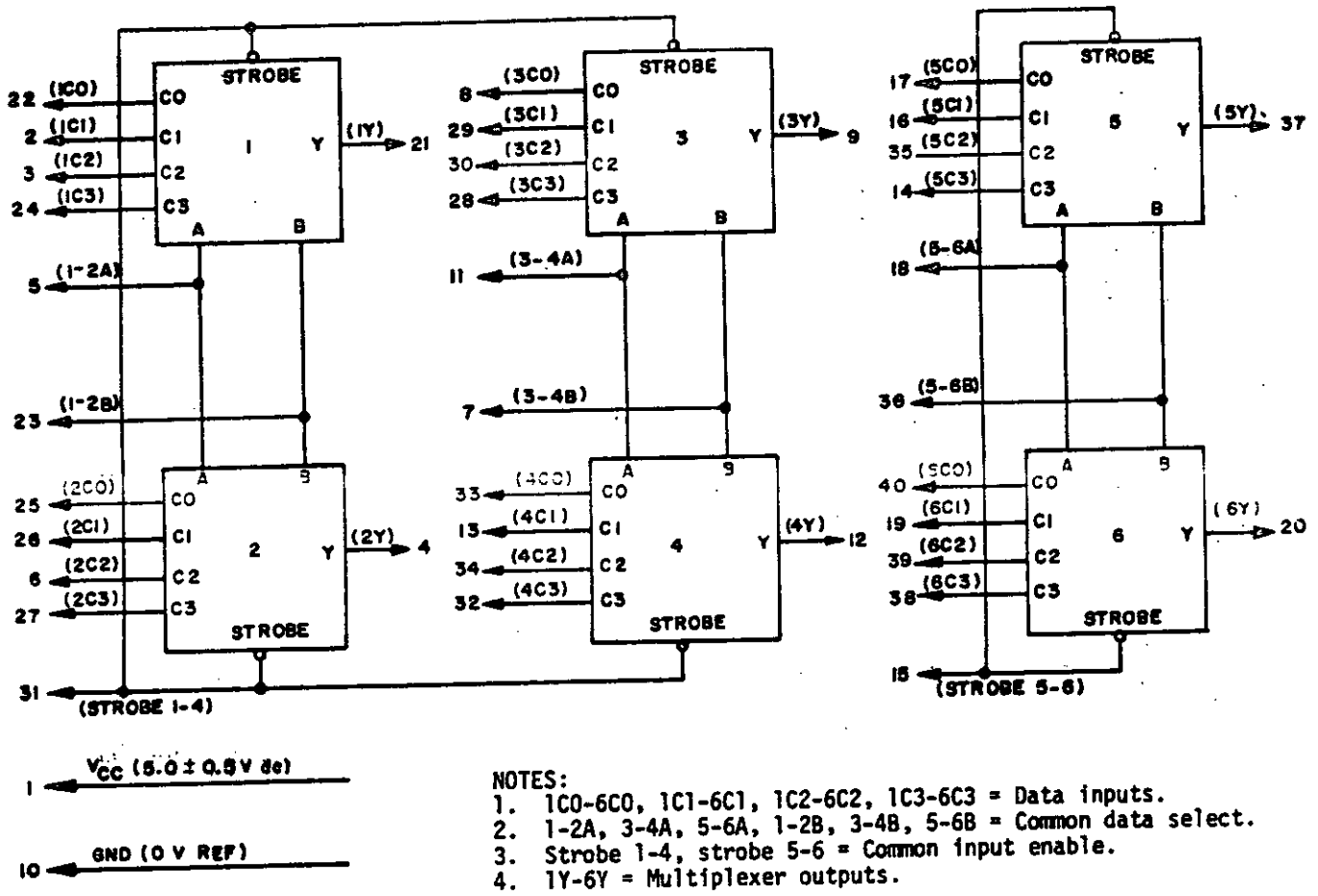
3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

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NOTES:
1. 1CO-6CO, 1C1-6C1, 1C2-6C2, 1C3-6C3 = Data inputs.
2. 1-2A, 3-4A, 5-6A, 1-2B, 3-4B, 5-6B = Common data select.
3. Strobe 1-4, strobe 5-6 = Common input enable.
4. 1Y-6Y = Multiplexer outputs.

* FIGURE 1. AHD module function diagram, multiplexer.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/ 3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B6	V_{OUT} (low loaded)	IV	S	0.01	1.00	0.01	1.00	0.01	1.00	0.01	1.00	V
C1-C30	Input breakdown current	V	S	0	1	0	1	0	1	0	1	mA
C31	Input breakdown current	V	S	0	4	0	4	0	4	0	4	mA
C32	Input breakdown current	V	S	0	2	0	2	0	2	0	2	mA
D1-D6	Output leakage current	VI	X	0	1	0	1	0	1	0	1	mA
E1-E6	Output short circuit current	VII	X	-40	-100	-40	-100	-40	-100	-40	-100	mA
F1-F128	Functional test	VIII	X	← Test vectors must produce output vectors as specified in table VIII. →								
G1-G32	Input clamp diode voltage	IX	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
H1-H6	Output voltage, low level	X	X	0	500	0	500	0	500	0	500	mV
J1-J6	Output voltage, high level	XI	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
K1-K30	Input current, high level	XII	X	0	50	0	50	0	50	0	50	µA
K31	Input current, high level	XII	X	0	200	0	200	0	200	0	200	µA

See footnotes at end of table.

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TABLE I. Module electrical requirements - continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
K32	Input current, high level	XII	X	0	100	0	100	0	100	0	100	µA
L1-L30	Input current, low level	XIII	X	-0.1	-2.2	-0.1	-2.2	-0.1	-2.2	-0.1	-2.2	mA
L31	Input current, low level	XIII	X	-0.1	-8.8	-0.1	-8.8	-0.1	-8.8	-0.1	-8.8	mA
L32	Input current, low level	XIII	X	-0.1	-4.4	-0.1	-4.4	-0.1	-4.4	-0.1	-4.4	mA
M	Power supply current, low level outputs	XIV	X	45	210	45	210	45	210	45	210	mA
N1-N6	Transition time (t_{TLH}), C_L = 50 pF	XV	S	2	15	2	15	2	18	2	18	ns
N7-N12	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	2	27	2	27	2	30	2	30	ns
P1-P6	Transition time (t_{THL}), C_L = 50 pF	XV	S	2	6	2	6	2	8	2	8	ns
P7-P12	Transition time (t_{THL}), C_L = 150 pF	XV	Q	2	11	2	11	2	13	2	13	ns
Q1-Q24	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	2	11	2	11	2	12	2	12	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - continued.

Test condition	Parameter	Ref test table	Test type 1/ -3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature. 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q25-Q36	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	2	20	2	20	2	21	2	21	ns
Q37-Q42	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	2.0	17.5	2.0	17.5	2	19	2	19	ns
Q43-Q66	Propagation delay time (t_{PLH}), C_L = 150 pF	XV	Q	4.0	14.5	4.0	14.5	4	16	4	16	ns
Q67-Q78	Propagation delay time (t_{PLH}), C_L = 150 pF	XV	Q	4.0	23.5	4.0	23.5	4	25	4	25	ns
Q79-Q84	Propagation delay time (t_{PLH}), C_L = 150 pF	XV	Q	4	20	4	20	4	22	4	22	ns
R1-R24	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	2.0	10.5	2.0	10.5	2	11	2	11	ns
R25-R36	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	2.0	19.5	2.0	19.5	2	20	2	20	ns
R37-R42	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	2	15	2	15	2.0	15.5	2.0	15.5	ns
R43-R66	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	4.0	13.5	4.0	13.5	4.0	14.5	4.0	14.5	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R67-R78	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	4	23	4	23	4	24	4	24	ns
R79-R84	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	4	18	4	18	4	19	4	19	ns
S1-S6	Output crosstalk, low level	XVI	Q	0	650	0	650	0	650	0	650	mV
T1-T6	Output crosstalk, high level	XVI	Q	0	400	0	400	0	400	0	400	mV
I'	Power supply transient amplitude	XVII	Q	0	400	0	400	0	400	0	400	mV
V	Power supply filter capacitance	XVIII	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
W1-W30	Input capacitance	XIX	Q	2	15	2	15	2	15	2	15	pF
W31	Input capacitance	XIX	Q	2	45	2	45	2	45	2	45	pF
W32	Input capacitance	XIX	Q	2	30	2	30	2	30	2	30	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A,B,C,D, or E are part of a sequence of tests, the order of testing shall be as follows: A,B,C,D,E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the oerating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition V, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through W shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/78-1 supersedes part number M28787/78. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation All outputs at low level	1,155 mW (outputs not loaded)
Supply voltage	5 V dc ±10 percent
Input voltage Low level High level	-0.5 V minimum, 0.8 V maximum 2.0 V minimum V_{CC} maximum
Input current, C0-C3, A, B Low level (V_{IL} = 0.0 V) (V_{IL} = 0.5 V) High level (V_{IH} = 2.5 V) (V_{IH} = 5.5 V)	 -2.2 mA maximum -2.0 mA maximum 50 µA maximum 1.0 mA maximum
Input current, STROBE 1-4 Low level (V_{IL} = 0.0 V) (V_{IL} = 0.5 V) High level (V_{IH} = 2.5 V) (V_{IH} = 5.5 V)	 -8.8 mA maximum -8.0 mA maximum 200 µA maximum 4.0 mA maximum
Input current, STROBE 5-6 Low level (V_{IL} = 0.0 V) (V_{IL} = 0.5 V) High level (V_{IH} = 2.5 V) (V_{IH} = 5.5 V)	 -4.4 mA maximum -4.0 mA maximum 100 µA maximum 2.0 mA maximum
Output voltage Low level (I_{OL} = 20 mA) High level (I_{OH} = -1.0 mA)	 0.5 V maximum 2.5 V minimum
Output drive current maximum capability Low level (V_{OL} = 0.5 V) High level (V_{OH} = 2.5 V)	 20 mA -1.0 mA

* 6.5 Module function. This module contains six programmable four-input multiplexers. Each multiplexer receives four input data lines (C0 through C3), two input data select lines (A and B), and an input enable line (STROBE). The input data line used is selected by the state of the data select inputs. This data is transferred to the output (Y) when the input enable (STROBE) is active (low). When it is not activated, the output is low regardless of all other inputs. The input enable (STROBE) is common to multiplexers 1 through 4, 5, and 6. The input data select lines (A and B) are common to multiplexers 1 and 2, 3 and 4, and to 5 and 6.

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* 6.5.1 Input to output relationship. Each multiplexer has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/-3/

Select inputs		Data inputs				Strobe	Output
B	A	C0	C1	C2	C3	STROBE	Y
X	X	X	X	X	X	H	L
L	L	L	X	X	X	L	L
L	L	H	X	X	X	L	H
L	H	X	L	X	X	L	L
L	H	X	H	X	X	L	H
H	L	X	X	L	X	L	L
H	L	X	X	H	X	L	H
H	H	X	X	X	L	L	L
H	H	X	X	X	H	L	H

1/ H = High-level.

2/ L = Low-level.

3/ X = Don't care condition.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

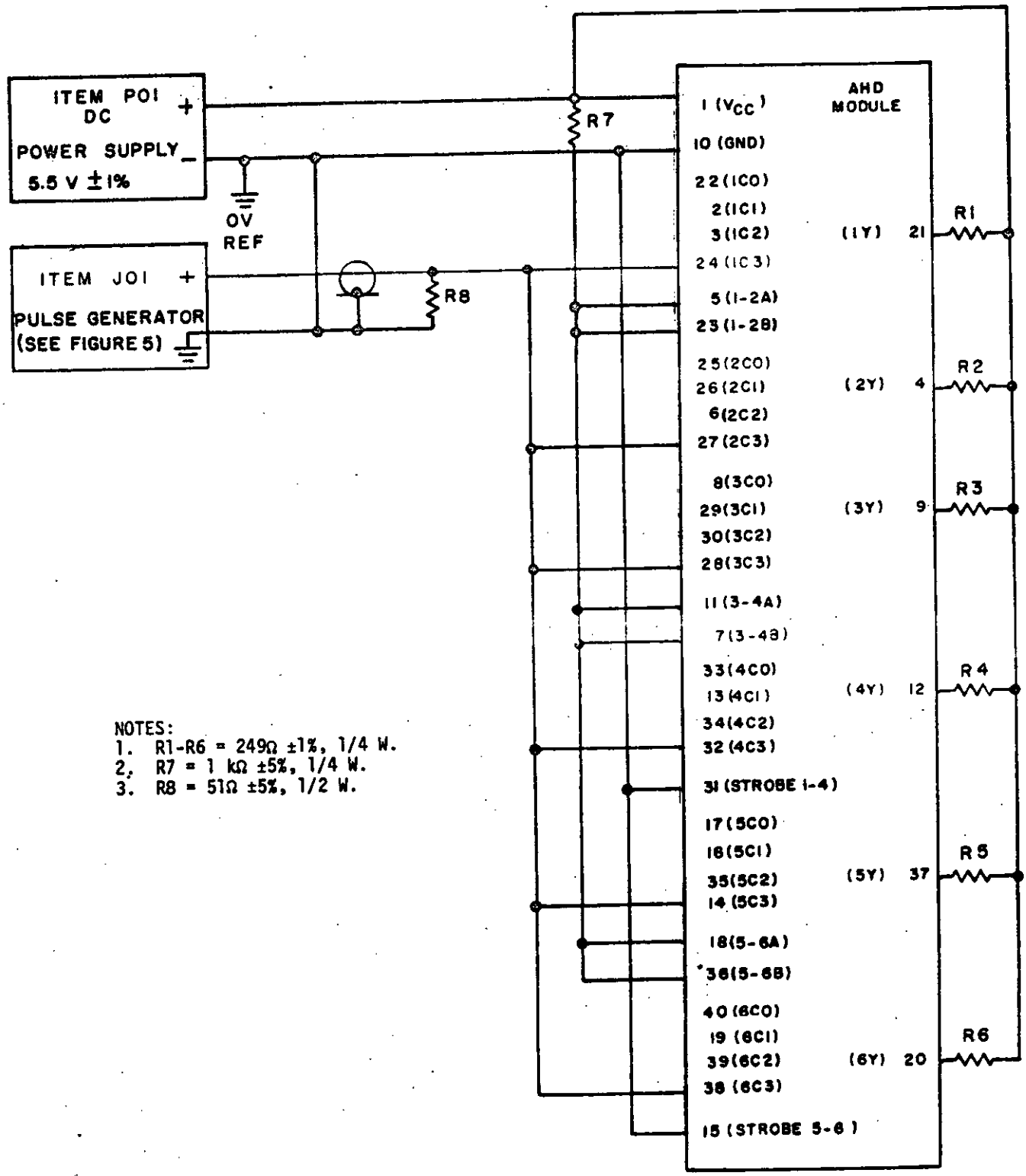
6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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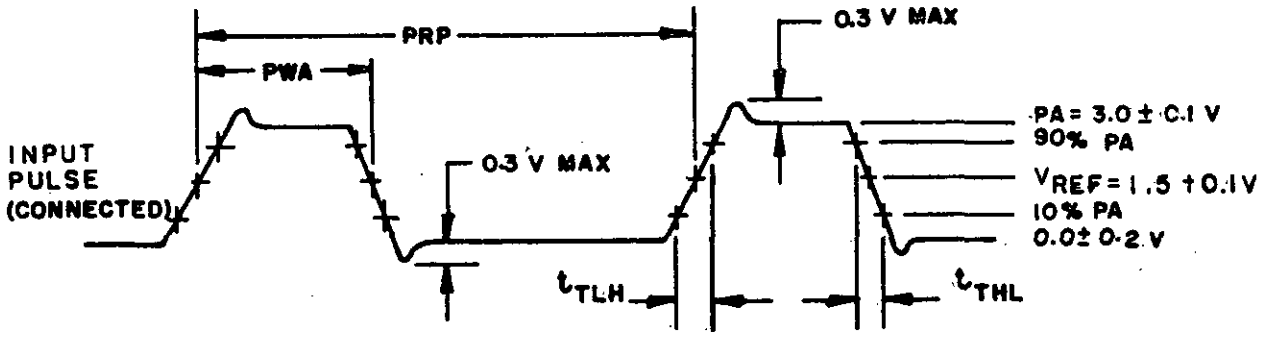
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NOTES:
1. R1-R6 = 249Ω ±1%, 1/4 W.
2. R7 = 1 kΩ ±5%, 1/4 W.
3. R8 = 51Ω ±5%, 1/2 W.

* FIGURE 2. Operating temperature and life test circuit.

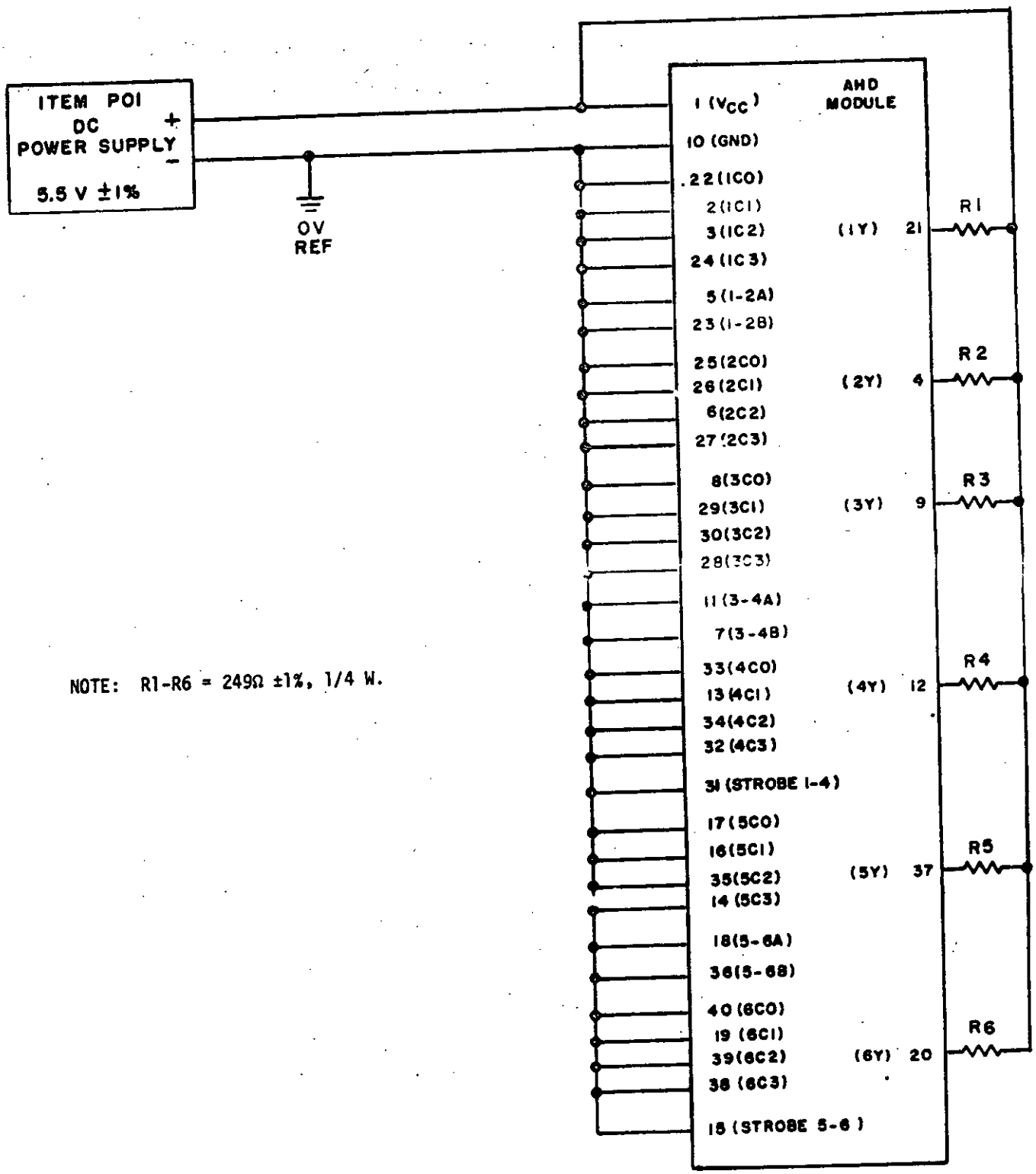
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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} \leq 10$ ns
PRP = 1,000 ±100 ns.
PWA = 500 ±50 ns.

* FIGURE 3. Operating temperature and life test circuit waveform.

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NOTE: R1-R6 = 249Ω ±1%, 1/4 W.

FIGURE 4. Component temperature test circuit.

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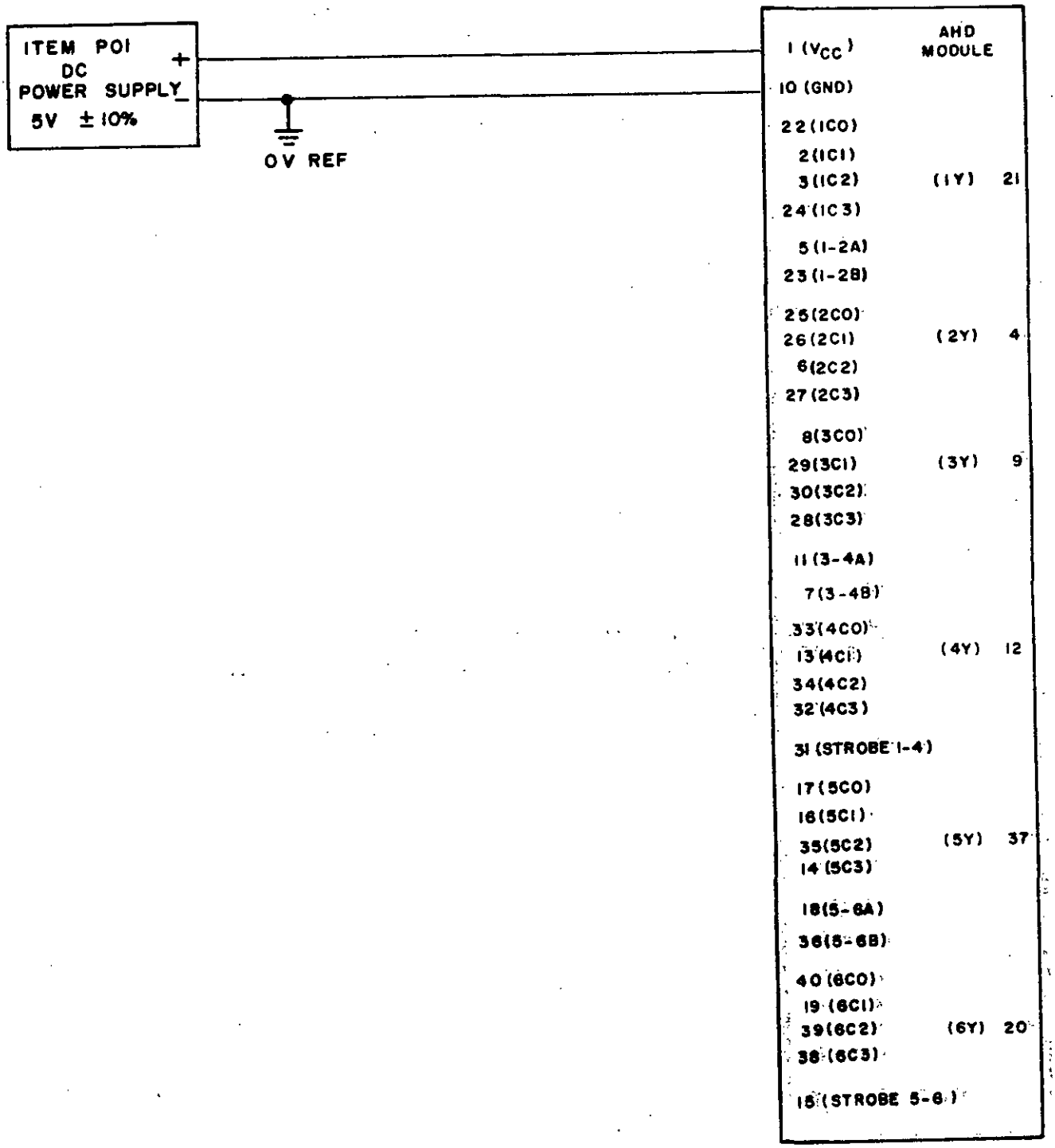


FIGURE 5. Vibration test circuit.

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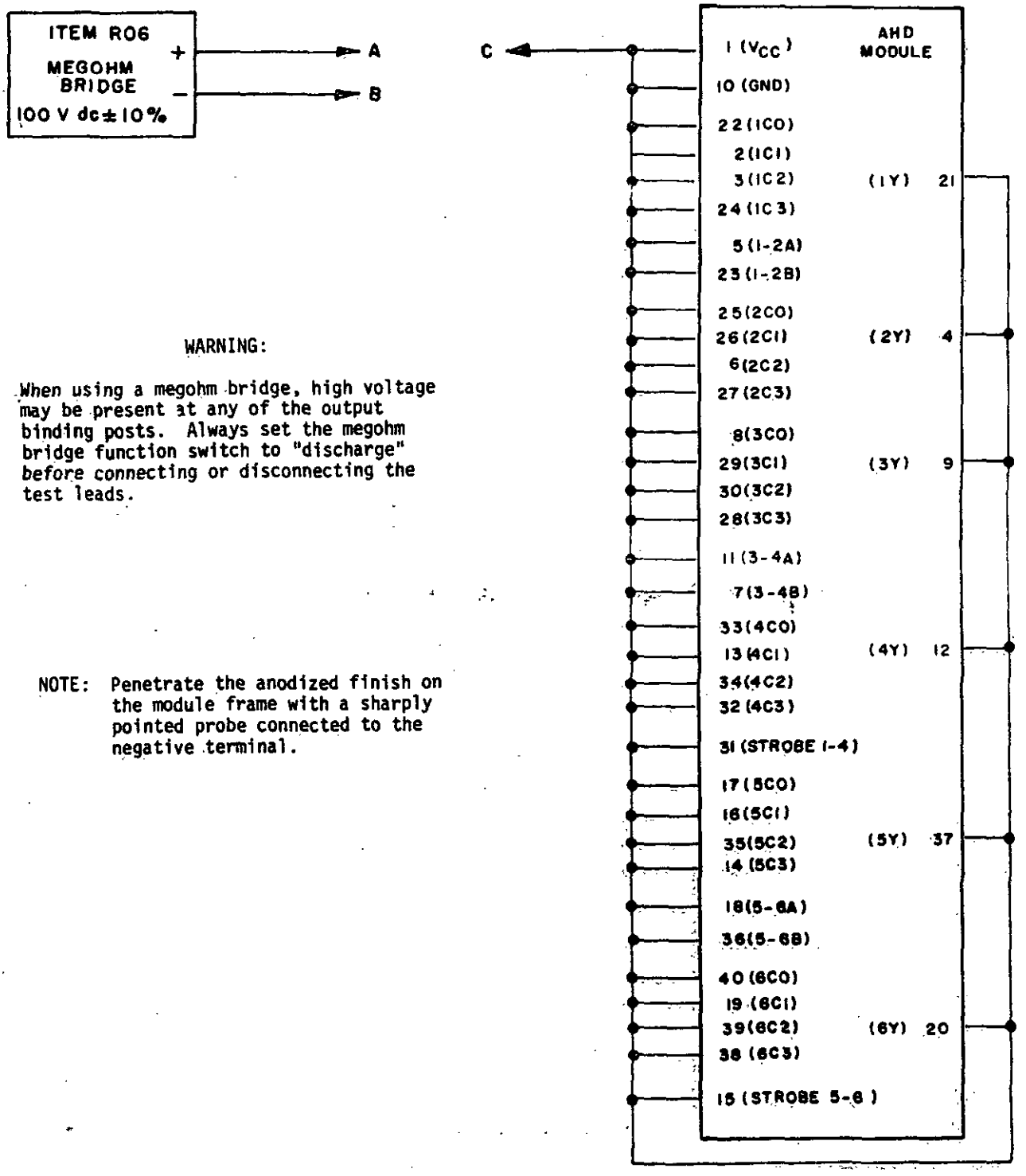
TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	6

TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
B1	1Y V_{OUT} (low loaded)	21	0.01	1.00	V	7
B2	2Y	4	↓	↓	↓	↓
B3	3Y	9				
B4	4Y	12				
B5	5Y	37				
B6	6Y	20				

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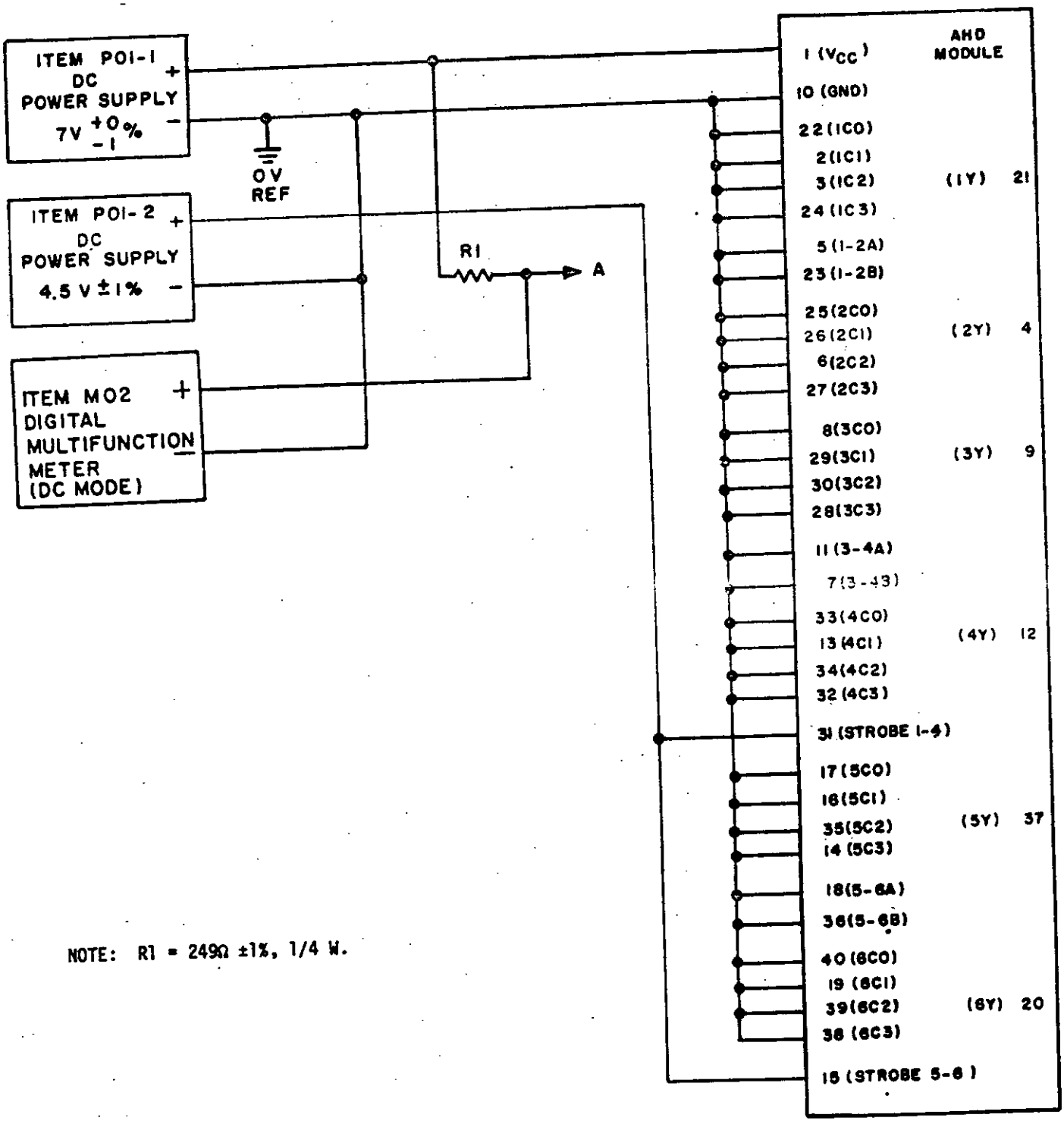
WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 6. Isolation test circuit for table III.

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NOTE: R1 = 249Ω ±1%, 1/4 W.

* FIGURE 7. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1CO Input breakdown current	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,23,24,25,26, 27,28,29,30,32,33,34,35, 36,38,39,40	22	0	1	mA	8
C2	1C1	3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33, 34,35,36,38,39,40	2				
C3	1C2	2,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33 34,35,36,38,39,40	3				
C4	1C3	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,25,26, 27,28,29,30,31,32,33,34, 35,36,38,39,40	24				
C5	2CO	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,26, 27,28,29,30,32,33,34,35, 36,38,39,40	25				
C6	2C1	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 27,28,29,30,31,32,33,34, 35,36,38,39,40	26				
C7	2C2	2,3,5,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33, 34,35,36,38,39,40	6				

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TABLE V. Input breakdown current test - continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C8	2C3 Input breakdown current	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,28,29,30,31,32,33,34, 35,36,38,39,40	27	0	1	mA	8
C9	3C0	2,3,5,6,7,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,32,33,34, 35,36,38,39,40	8				
C10	3C1	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,30,31,32,33,34, 35,36,38,39,40	29				
C11	3C2	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,31,32,33,34, 35,36,38,39,40	30				
C12	3C3	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,29,30,31,32,33,34, 35,36,38,39,40	28				
C13	4C0	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,32,34,35, 36,38,39,40	33				
C14	4C1	2,3,5,6,7,8,11,14,15,16, 17,18,19,22,23,24,25,26, 27,28,29,30,31,32,33,34, 35,36,38,39,40	13				

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TABLE V. Input breakdown current test - continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C15	4C2 Input breakdown current	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33, 35,36,38,39,40	34	0	1	mA	8
C16	4C3	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,33,34, 35,36,38,39,40	32				
C17	5C0	2,3,5,6,7,8,11,13,14,16, 18,19,22,23,24,25,26,27, 28,29,30,31,32,33,34,35, 36,38,39,40	17				
C18	5C1	2,3,5,6,7,8,11,13,14,15, 17,18,19,22,23,24,25,26, 27,28,29,30,31,32,33,34, 35,36,38,39,40	16				
C19	5C2	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33, 34,36,38,39,40	35				
C20	5C3	2,3,5,6,7,8,11,13,15,16, 17,18,19,22,23,24,25,26, 27,28,29,30,31,32,33,34, 35,36,38,39,40	14				
C21	6C0	2,3,5,6,7,8,11,13,14,16, 17,18,19,22,23,24,25,26, 27,28,29,30,31,32,33,34, 35,36,38,39	40				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:57Z
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TABLE V. Input breakdown current test - continued.

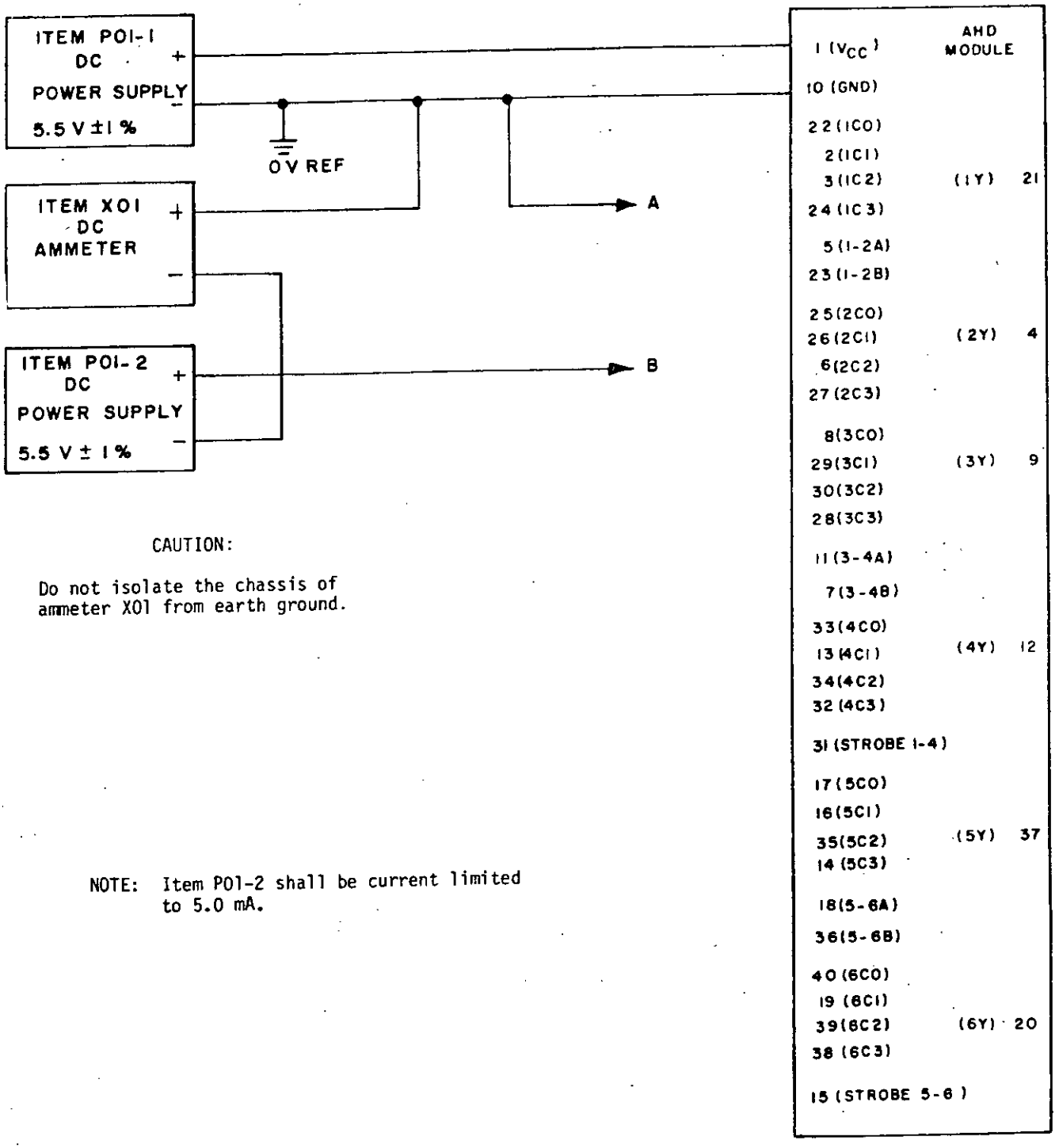
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C22	6C1 Input breakdown current	2,3,5,6,7,8,11,13,14,15, 16,17,18,22,23,24,25,26, 27,28,29,30,31,32,33,34, 35,36,38,39,40	19	0	1	mA	8
C23	6C2	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33, 34,35,36,38,40	39				
C24	6C3	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33, 34,35,36,39,40	38				
C25	1-2A	2,3,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33, 34,35,36,38,39,40	5				
C26	1-2B	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,24,25.26, 27,28,29,30,31,32,33,34, 35,36,38,39,40	23				
C27	3-4A	2,3,5,6,7,8,13,14,15,16, 17,18,19,22,23,24,25,26, 27,28,29,30,31,32,33,34, 35,36,38,39,40	11				
C28	3-4B	2,3,5,6,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33, 34,35,36,38,39,40	7				

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TABLE V. Input breakdown current test - continued.

Test condition	Parameter		Connect: A to pin	Connect: B to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
C29	5-6A	Input breakdown current	2,3,5,6,7,8,11,13,14,15, 16,17,19,22,23,24,25,26, 27,28,29,30,31,32,33,34, 35,36,38,39,40	18	0	1	mA	8
C30	5-6B		2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33, 34,35,38,39,40	36		1		
C31	STROBE 1-4		2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,32,33,34, 35,36,38,39,40	31		4		
C32	STROBE 5-6		2,3,5,6,7,8,11,13,14,16, 17,18,19,22,23,24,25,26, 27,28,29,30,31,32,33,34, 35,36,38,39,40	15		2		

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CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

NOTE: Item POI-2 shall be current limited to 5.0 mA.

* FIGURE 8. Input breakdown current test circuit for table V.

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TABLE VI. Output leakage current test.

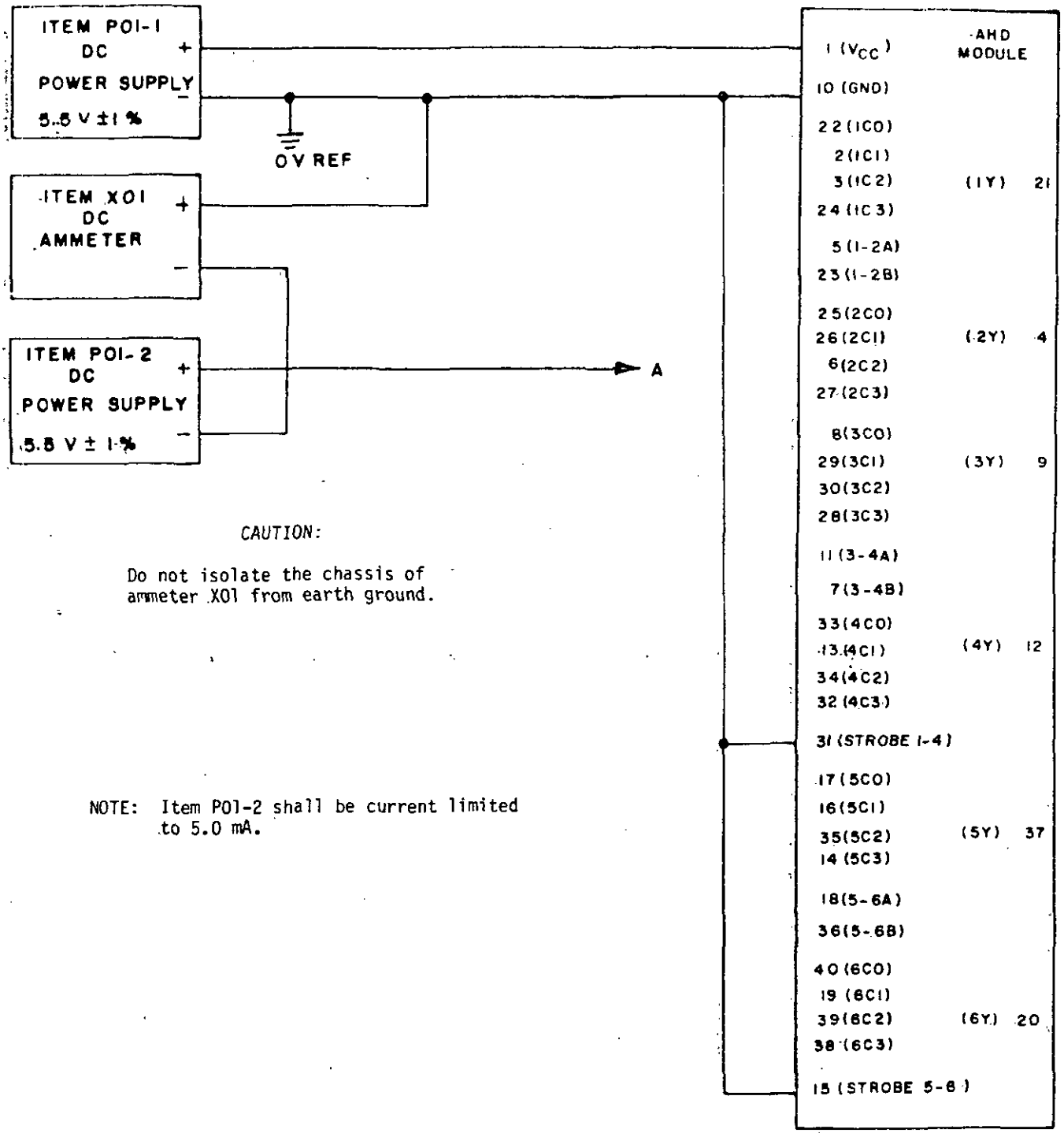
Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D1	1Y	Output leakage current	21	0	1	mA	9
D2	2Y	↓	4	↓	↓	↓	↓
D3	3Y		9				
D4	4Y		12				
D5	5Y		37				
D6	6Y		20				

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TABLE VII. Output short circuit current test.

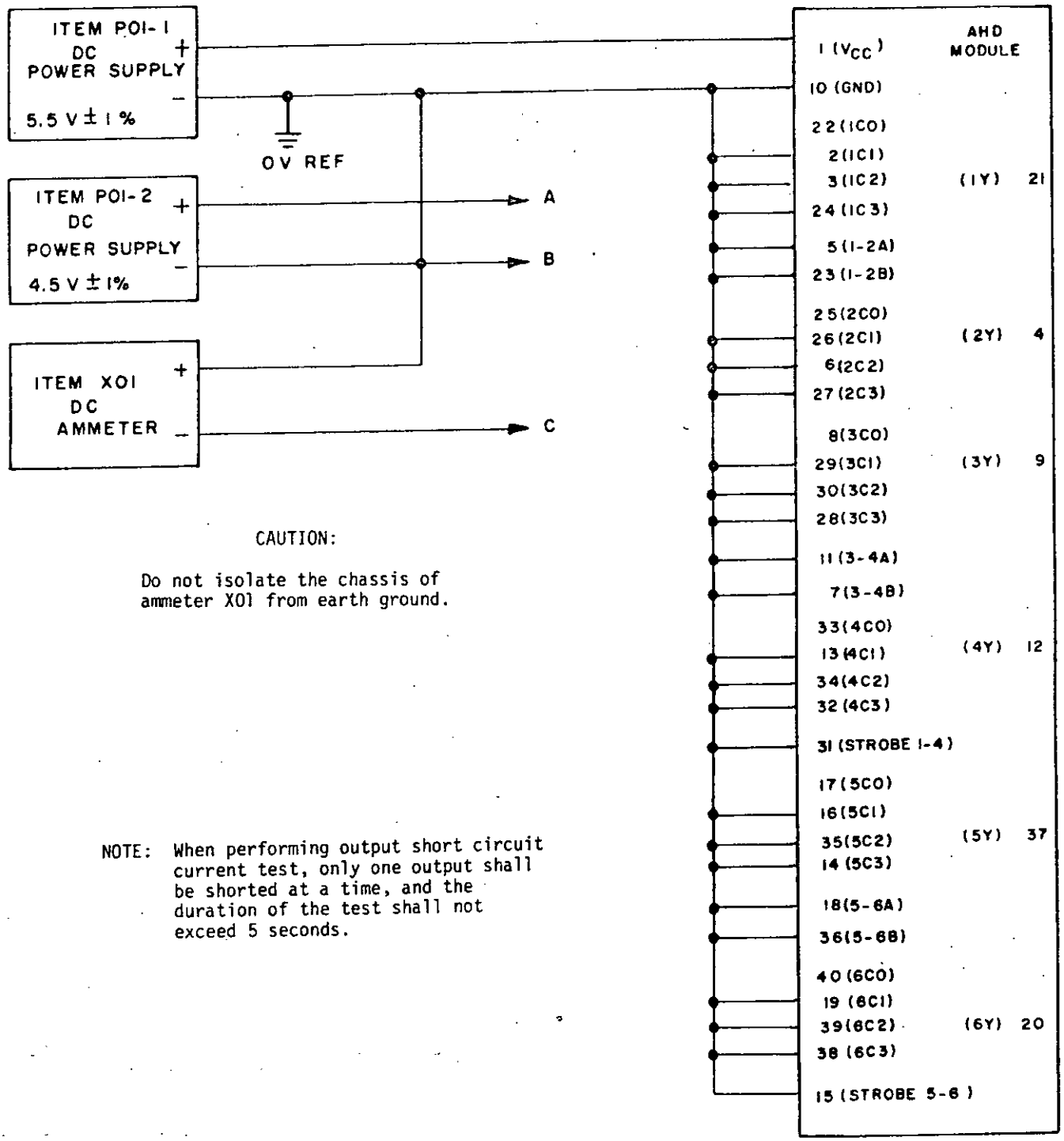
Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
E1	1Y	Output short circuit current	22	8,17,25, 33,40	21	-40	-100	mA	10
E2	2Y	↓	25	8,17,22, 33,40	4	↓	↓	↓	↓
E3	3Y	↓	8	17,22,25, 33,40	9	↓	↓	↓	↓
E4	4Y	↓	33	8,17,22, 25,40	12	↓	↓	↓	↓
E5	5Y	↓	17	8,22,25, 33,40	37	↓	↓	↓	↓
E6	6Y	↓	40	8,17,22, 25,33	20	↓	↓	↓	↓

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* FIGURE 9. Output leakage current test circuit for table VI.

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE: When performing output short circuit current test, only one output shall be shorted at a time, and the duration of the test shall not exceed 5 seconds.

FIGURE 10. Output short circuit current test circuit for table VII.

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TABLE VIII. Functional test.

Test condition	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	4Y	5Y	6Y	Min	Max		
F1	Functional	Open	2,3,5,6,7,8,11,13,14, 15,16,17,18,19,22,23, 24,25,26,27,28,29,30, 31,32,33,34,35,36,38, 39,40	2/ 3/	L	L	L	L	L	L	1/			11
F2		15,31	2,3,5,6,7,8,11,13,14, 16,17,18,19,22,23,24, 25,26,27,28,29,30,32, 33,34,35,36,38,39,40		L	L	L	L	L	L				
F3		14,24,27,28,32,38	2,3,5,6,7,8,11,13,15, 16,17,18,19,22,23,25, 26,29,30,31,33,34,35, 36,39,40		L	L	L	L	L	L				
F4		14,15,24,27,28,31,32, 38	2,3,5,6,7,8,11,13,16, 17,18,19,22,23,25,26, 29,30,33,34,35,36,39, 40		L	L	L	L	L	L				
F5		3,6,30,34,35,39	2,5,7,8,11,13,14,15, 16,17,18,19,22,23,24, 25,26,27,28,29,31,32, 33,36,38,40		L	L	L	L	L	L				
F6		3,6,15,30,31,34,35,39	2,5,7,8,11,13,14,16, 17,18,19,22,23,24,25, 26,27,28,29,32,33,36, 38,40		L	L	L	L	L	L				
F7		3,6,14,24,27,28,30, 32,34,35,38,39	2,5,7,8,11,13,15,16, 17,18,19,22,23,25,26, 29,31,33,36,40		L	L	L	L	L	L				

See footnotes at end of table.

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TABLE VIII. Functional test - continued.

Test condition	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	4Y	5Y	6Y	Min	Max		
F8	Functional	3,6,14,15,24,27,28, 30,31,32,34,35,38,39	2,5,7,8,11,13,15,16, 17,18,19,22,23,25,26, 29,33,36,40	2/ 3/	L	L	L	L	L	L		1/		11
F9		2,13,16,19,26,29	3,5,6,7,8,11,14,15, 17,18,22,23,24,25,27, 28,30,31,32,33,34,35, 36,38,39,40		L	L	L	L	L	L				
F10		2,13,15,16,19,26,29, 31	3,5,6,7,8,11,14,17 18,22,23,24,25,27,28, 30,32,33,34,35,36,38, 39,40		L	L	L	L	L	L				
F11		2,13,14,16,19,24,26, 27,28,29,32,38	3,5,6,7,8,11,15,17, 18,22,23,25,30,31,33, 34,35,36,39,40		L	L	L	L	L	L				
F12		2,13,14,15,16,19,24, 26,27,28,29,31,32,38	5,6,7,8,11,13,17,18, 22,23,25,30,33,34,35, 36,39,40		L	L	L	L	L	L				
F13		2,3,6,13,16,19,26,29, 30,34,35,39	5,7,8,11,14,15,17,18, 22,23,24,25,27,28,31, 32,33,36,38,40		L	L	L	L	L	L				
F14		2,3,6,13,15,16,19,26, 29,30,31,34,35,39	5,7,8,11,14,17,18,22, 23,24,25,27,28,32,33, 36,38,40		L	L	L	L	L	L				
F15		2,3,6,13,14,16,19,24, 26,27,28,29,30,32,34, 35,38,39	5,7,8,11,15,17,18,22, 23,25,31,33,36,40		L	L	L	L	L	L				
F16		2,3,6,13,14,15,16,19, 24,26,27,28,29,30,31, 32,34,35,38,39	5,7,8,11,17,18,22,23, 25,33,36,40		L	L	L	L	L	L				

See footnotes at end of table.

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TABLE VIII. Functional test - continued.

Test condition	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	4Y	5Y	6Y	Min	Max		
F17	Functional	8,17,22,25,33,40	2,3,5,6,7,11,13,14, 15,16,18,19,23,24,26, 27,28,29,30,31,32,34, 35,36,38,39	2/ 3/	H	H	H	H	H	H		1/		11
F18		8,15,17,22,25,31,33, 40	2,3,5,6,7,11,13,14, 16,18,19,23,24,26,27, 28,29,30,32,34,35,36, 38,39		L	L	L	L	L	L				
F19		8,14,17,22,24,25,27, 28,32,33,38,40	2,3,5,6,7,11,13,15, 16,18,19,23,26,29,30, 31,34,35,36,39		H	H	H	H	H	H				
F20		8,14,15,17,22,24,25, 27,28,31,32,33,38,40	2,3,5,6,7,11,13,16, 18,19,23,26,29,30,34, 35,36,39		L	L	L	L	L	L				
F21		3,6,8,17,22,25,30,33, 34,35,39,40	2,5,7,11,13,14,15,16, 18,19,23,24,26,27,28, 29,31,32,36,38		H	H	H	H	H	H				
F22		3,6,8,15,17,22,25,30, 31,33,34,35,39,40	2,5,7,11,13,14,16,18, 19,23,24,26,27,28,29, 32,36,38		L	L	L	L	L	L				
F23		3,6,8,14,17,22,24,25, 27,28,30,32,33,34,35, 38,39,40	2,5,7,11,13,15,16,18, 19,23,26,29,31,36		H	H	H	H	H	H				
F24		3,6,8,14,15,17,22,24, 25,27,28,30,31,32,33, 34,35,38,39,40	2,5,7,11,13,16,18,19, 23,26,29,36		L	L	L	L	L	L				
F25		2,8,13,16,17,19,22, 25,26,29,33,40	3,5,6,7,11,14,15,18, 23,24,27,28,30,31,32, 34,35,36,38,39		H	H	H	H	H	H				

See footnotes at end of table.

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TABLE VIII. Functional test - continued.

Test condition	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	4Y	5Y	6Y	Min	Max		
F26	Functional	2,8,13,15,16,17,19, 22,25,26,29,31,33,40	3,5,6,7,11,14,18,23, 24,27,28,30,32,34,35, 36,38,39	2/ 3/	L	L	L	L	L	L		1/		11
F27		2,8,13,14,16,17,19, 22,24,25,26,27,28,29, 32,33,38,40	3,5,6,7,11,15,18,23, 30,31,34,35,36,39		H	H	H	H	H	H				
F28		2,8,13,14,15,16,17, 19,22,24,25,26,27,28, 29,31,32,33,38,40	3,5,6,7,11,18,23,30, 34,35,36,39		L	L	L	L	L	L				
F29		2,3,6,8,13,16,17,19, 22,25,26,29,30,33,34, 35,39,40	5,7,11,14,15,18,23, 24,27,28,31,32,36,38		H	H	H	H	H	H				
F30		2,3,6,8,13,15,16,17, 19,22,25,26,29,30,31, 33,34,35,39,40	5,7,11,14,18,23,24, 27,28,32,36,38		L	L	L	L	L	L				
F31		2,3,6,8,13,14,16,17, 19,22,24,25,26,27,28, 29,30,32,33,34,35,38, 39,40	5,7,11,15,18,23,31, 36		H	H	H	H	H	H				
F32		2,3,6,8,13,14,15,16, 17,19,22,24,25,26,27, 28,29,30,31,32,33,34, 35,38,39,40	5,7,11,18,23,36		L	L	L	L	L	L				
F33		5,11,18	2,3,6,7,8,13,14,15, 16,17,19,22,23,24,25, 26,27,28,29,30,31,32, 33,34,35,36,38,39,40		L	L	L	L	L	L				

See footnotes at end of table.

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TABLE VIII. Functional test - continued.

Test condition	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	4Y	5Y	6Y	Min	Max		
F34	Functional	5,11,15,18,31	2,3,6,7,8,13,14,16, 17,19,22,23,24,25,26, 27,28,29,30,32,33,34, 35,36,38,39,40	2/ 3/	L	L	L	L	L	L		1/		11
F35		5,11,14,18,24,27,28, 32,38	2,3,6,7,8,13,15,16, 17,19,22,23,25,26,29, 30,31,33,34,35,36,39, 40		L	L	L	L	L	L				
F36		5,11,14,15,18,24,27, 28,31,32,38	2,3,6,7,8,13,16,17, 19,22,23,25,26,29,30, 33,34,35,36,39,40		L	L	L	L	L	L				
F37		3,5,6,11,18,30,34,35, 39	2,7,8,13,14,15,16,17, 19,22,23,24,25,26,27, 28,29,31,32,33,36,38, 40		L	L	L	L	L	L				
F38		3,5,6,11,15,18,30,31, 34,35,39	2,7,8,13,14,16,17,19, 22,23,24,25,26,27,28, 29,32,33,36,38,40		L	L	L	L	L	L				
F39		3,5,6,11,14,18,24,27, 28,30,32,34,35,38,39	2,7,8,13,15,16,17,19, 22,23,25,26,29,31,33, 36,40		L	L	L	L	L	L				
F40		3,5,6,11,14,15,18,24, 27,28,30,31,32,34,35, 38,39	2,7,8,13,16,17,19,22, 23,25,26,29,33,36,40		L	L	L	L	L	L				
F41		2,5,11,13,16,18,19, 26,29	3,6,7,8,14,15,17,22, 23,24,25,27,28,30,31, 32,33,34,35,36,38,39, 40		H	H	H	H	H	H				

See footnotes at end of table.

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TABLE VIII. Functional test - continued.

Test condition	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	4Y	5Y	6Y	Min	Max		
F42	Functional	2,5,11,13,15,16,18, 19,26,29,31	3,6,7,8,14,17,22,23, 24,25,27,28,30,32,33, 34,35,36,38,39,40	2/ 3/	L	L	L	L	L	L	1/			11
F43		2,5,11,13,14,16,18, 19,24,26,27,28,29,32, 38	3,6,7,8,15,17,22,23, 25,30,31,33,34,35,36, 39,40		H	H	H	H	H	H				
F44		2,5,11,13,14,15,16, 18,19,24,26,27,28,29, 31,32,38	3,6,7,8,17,22,23,25, 30,33,34,35,36,39,40		L	L	L	L	L	L				
F45		2,3,5,6,11,13,16,18, 19,26,29,30,34,35,39	7,8,14,15,17,22,23, 24,25,27,28,31,32,33, 36,38,40		H	H	H	H	H	H				
F46		2,3,5,6,11,13,15,16, 18,19,26,29,30,31,34, 35,39	7,8,14,17,22,23,24, 25,27,28,32,33,36,38, 40		L	L	L	L	L	L				
F47		2,3,5,6,11,13,14,16, 18,19,24,26,27,28,29, 30,32,34,35,38,39	7,8,15,17,22,23,25, 31,33,36,40		H	H	H	H	H	H				
F48		2,3,5,6,11,13,14,15, 16,18,19,24,26,27,28, 19,30,31,32,34,35,38, 39	7,8,17,22,23,25,33, 36,40		L	L	L	L	L	L				
F49		5,8,11,17,18,22,25, 33,40	2,3,6,7,13,14,15,16, 19,23,24,26,27,28,29, 30,31,32,34,35,36,38, 39		L	L	L	L	L	L				
F50		5,8,11,15,17,18,22, 25,31,33,40	2,3,6,7,13,14,16,19, 23,24,26,27,28,29,30, 32,34,35,36,38,39		L	L	L	L	L	L				

See footnotes at end of table.

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TABLE VIII. Functional test - continued.

Test condition	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	4Y	5Y	6Y	Min	Max		
F51	Functional	5,8,11,14,17,18,22, 24,25,27,28,32,33,38, 40	2,3,6,7,13,15,16,19, 23,26,29,30,31,34,35, 36,39	2/ 3/	L	L	L	L	L	L		1/		11
F52		5,8,11,14,15,17,18, 22,24,25,27,28,31,32, 33,38,40	2,3,6,7,13,16,19,23, 26,29,30,34,35,36,39		L	L	L	L	L	L				
F53		3,5,6,8,11,17,18,22, 25,30,33,34,35,39,40	2,7,13,14,15,16,19, 23,24,26,27,28,29,31, 32,36,38		L	L	L	L	L	L				
F54		3,5,6,8,11,15,17,18, 22,25,30,31,33,34,35, 39,40	2,7,13,14,16,19,23, 24,26,27,28,29,32,36, 38		L	L	L	L	L	L				
F55		3,5,6,8,11,14,17,18, 22,24,25,27,28,30,32, 33,34,35,38,39,40	2,7,13,15,16,19,23, 26,29,31,36		L	L	L	L	L	L				
F56		3,5,6,8,11,14,15,17, 18,22,24,25,27,28,30, 31,32,33,34,35,38,39, 40	2,7,13,16,19,23,26, 29,36		L	L	L	L	L	L				
F57		2,5,8,11,13,16,17,18, 19,22,25,26,29,33,40	3,6,7,14,15,23,24,27, 28,30,31,32,34,35,36, 38,39		H	H	H	H	H	H				
F58		2,5,8,11,13,15,16,17, 18,19,22,25,26,29,31, 33,40	3,6,7,14,23,24,27,28, 30,32,34,35,36,38,39		L	L	L	L	L	L				
F59		2,5,8,11,13,14,16,17, 18,19,22,24,25,26,27, 28,29,32,33,38,40	3,6,7,15,23,30,31,34, 35,36,39		H	H	H	H	H	H				

See footnotes at end of table.

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TABLE VIII. Functional test - continued.

Test condition	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	4Y	5Y	6Y	Min	Max		
F60	Functional	2,5,8,11,13,14,15,16, 17,18,19,22,24,25,26, 27,28,29,31,32,33,38, 40	3,6,7,23,30,34,35,36, 39	2/ 3/	L	L	L	L	L	L	1/			11
F61		2,3,5,6,8,11,13,16, 17,18,19,22,25,26,29, 30,33,34,35,39,40	7,14,15,23,24,27,28, 31,32,36,38		H	H	H	H	H	H				
F62		2,3,5,6,8,11,13,15, 16,17,18,19,22,25,26, 29,30,31,33,34,35,39, 40	7,14,23,24,27,28,32, 36,38		L	L	L	L	L	L				
F63		2,3,5,6,8,11,13,14, 16,17,18,19,22,24,25, 26,27,28,29,30,32,33, 34,35,38,39,40	7,15,23,31,36		H	H	H	H	H	H				
F64		2,3,5,6,8,11,13,14, 15,16,17,18,19,22,24, 25,26,27,28,29,30,31, 32,33,34,35,38,39,40	7,23,36		L	L	L	L	L	L				
F65		7,23,36	2,3,5,6,8,11,13,14, 15,16,17,18,19,22,24, 25,26,27,28,29,30,31, 32,33,34,35,38,39,40		L	L	L	L	L	L				
F66		7,15,23,31,36	2,3,5,6,8,11,13,14, 16,17,18,19,22,24,25, 26,27,28,29,30,32,33, 34,35,38,39,40		L	L	L	L	L	L				
F67		7,14,23,24,27,28,32, 36,38	2,3,5,6,8,11,13,15, 16,17,18,19,22,25,26, 29,30,31,33,34,35,39, 40		L	L	L	L	L	L				

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TABLE VIII. Functional test - continued.

Test condition	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	4Y	5Y	6Y	Min	Max		
F68	Functional	7,14,15,23,24,27,28,31,32,36,38	2,3,5,6,8,11,13,16,17,18,19,22,25,26,29,30,33,34,35,39,40	2/ 3/	L	L	L	L	L	L		1/		11
F69		3,6,7,23,30,34,35,36,39	2,5,8,11,13,14,15,16,17,18,19,22,24,25,26,27,28,29,31,32,33,38,40		H	H	H	H	H	H				
F70		3,6,7,15,23,30,31,34,35,36,39	2,5,8,11,13,14,16,17,18,19,22,24,25,26,27,28,29,32,33,38,40		L	L	L	L	L	L				
F71		3,6,7,14,23,24,27,28,30,32,34,35,36,38,39	2,5,8,11,13,15,16,17,18,19,22,25,26,29,31,33,40		H	H	H	H	H	H				
F72		3,6,7,14,15,23,24,27,28,30,31,32,34,35,36,38,39	2,5,8,11,13,16,17,18,19,22,25,26,29,33,40		L	L	L	L	L	L				
F73		2,7,13,16,19,23,26,29,36	3,5,6,8,11,14,15,17,18,22,24,25,27,28,30,31,32,33,34,35,38,39,40		L	L	L	L	L	L				
F74		2,7,13,15,16,19,23,26,29,31,36	3,5,6,8,11,14,17,18,22,24,25,27,28,30,32,33,34,35,38,39,40		L	L	L	L	L	L				
F75		2,7,13,14,16,19,23,24,26,27,28,29,32,36,38	3,5,6,8,11,15,17,18,22,25,30,31,33,34,35,39,40		L	L	L	L	L	L				
F76		2,7,13,14,15,16,19,23,24,26,27,28,29,31,32,36,38	3,5,6,8,11,17,18,22,25,30,33,34,35,39,40		L	L	L	L	L	L				

See footnotes at end of table.

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TABLE VIII. Functional test - continued.

Test condition	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	4Y	5Y	6Y	Min	Max		
F77	Functional	2,3,6,7,13,16,19,23, 26,29,30,34,35,36,39	5,8,11,14,15,17,18, 22,24,25,27,28,31,32, 33,38,40	2/ 3/	H	H	H	H	H	H	1/			11
F78		2,3,6,7,13,15,16,19, 23,26,29,30,31,34,35, 36,39	5,8,11,14,17,18,22, 24,25,27,28,32,33,38, 40		L	L	L	L	L	L				
F79		2,3,6,7,13,14,16,19, 23,24,26,27,28,29,30, 32,34,35,36,38,39	5,8,11,15,17,18,22, 25,31,33,40		H	H	H	H	H	H				
F80		2,3,6,7,13,14,15,16, 19,23,24,26,27,28,29, 30,31,32,34,35,36,38, 39	5,8,11,17,18,22,25, 33,40		L	L	L	L	L	L				
F81		7,8,17,22,23,25,33, 36,40	2,3,5,6,11,13,14,15, 16,18,19,24,26,27,28, 29,30,31,32,34,35,38, 39		L	L	L	L	L	L				
F82		7,8,15,17,22,23,25, 31,33,36,40	2,3,5,6,11,13,14,16, 18,19,24,26,27,28,29, 30,32,34,35,38,39		L	L	L	L	L	L				
F83		7,8,14,17,22,23,24, 25,27,28,32,33,36,38, 40	2,3,5,6,11,13,15,16, 18,19,26,29,30,31,34, 35,39		L	L	L	L	L	L				
F84		7,8,14,15,17,22,23, 24,25,27,28,31,32,33, 36,38,40	2,3,5,6,11,13,16,18, 19,26,29,30,34,35,39		L	L	L	L	L	L				
F85		3,6,7,8,17,22,23,25, 30,33,34,35,36,39,40	2,5,11,13,14,15,16, 18,19,24,26,27,28,29, 31,32,38		H	H	H	H	H	H				

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TABLE VIII. Functional test - continued.

Test condition	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	4Y	5Y	6Y	Min	Max		
F86	Functional	3,6,7,8,15,17,22,23, 25,30,31,33,34,35,36, 39,40	2,5,11,13,14,16,18, 19,24,26,27,28,29,32, 38	2/ 3/	L	L	L	L	L	L	1/			11
F87		3,6,7,8,14,17,22,23, 24,25,27,28,30,32,33, 34,35,36,38,39,40	2,5,11,13,15,16,18, 19,26,29,31		H	H	H	H	H	H				
F88		3,6,7,8,14,15,17,22, 23,24,25,27,28,30,31, 32,33,34,35,36,38,39, 40	2,5,11,13,16,18,19, 26,29		L	L	L	L	L	L				
F89		2,7,8,13,16,17,19,22, 23,25,26,29,33,36,40	3,5,6,11,14,15,18,24, 27,28,30,31,32,34,35, 38,39		L	L	L	L	L	L				
F90		2,7,8,13,15,16,17,19, 22,23,25,26,29,31,33, 36,40	3,5,6,11,14,18,24,27, 28,30,32,34,35,38,39		L	L	L	L	L	L				
F91		2,7,8,13,14,16,17,19, 22,23,24,25,26,27,28, 29,32,33,36,38,40	3,5,6,11,15,18,30,31, 34,35,39		L	L	L	L	L	L				
F92		2,7,8,13,14,15,16,17, 19,22,23,24,25,26,27, 28,29,31,32,33,36,38, 40	3,5,6,11,18,30,34,35, 39		L	L	L	L	L	L				
F93		2,3,6,7,8,13,16,17, 19,22,23,25,26,29,30, 33,34,35,36,39,40	5,11,14,15,18,24,27, 28,31,32,38		H	H	H	H	H	H				
F94		2,3,6,7,8,13,15,16, 17,19,22,23,25,26,29, 30,31,33,34,35,36,39, 40	5,11,14,18,24,27,28, 32,38		L	L	L	L	L	L				

See footnotes at end of table.

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TABLE VIII. Functional test - continued.

Test condition	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	4Y	5Y	6Y	Min	Max		
F95	Functional	2,3,6,7,8,13,14,16, 17,19,22,23,24,25,26, 27,28,29,30,32,33,34, 35,36,38,39,40	5,11,15,18,31	2/ 3/	H	H	H	H	H	H	1/			11
F96		2,3,6,7,8,13,14,15, 16,17,19,22,23,24,25, 26,27,28,29,30,31,32, 33,34,35,36,38,39,40	5,11,18		L	L	L	L	L	L				
F97		5,7,11,18,23,36	2,3,6,8,13,14,15,16, 17,19,22,24,25,26,27, 28,29,30,31,32,33,34, 35,38,39,40		L	L	L	L	L	L				
F98		5,7,11,15,18,23,31, 36	2,3,6,8,13,14,16,17, 19,22,24,25,26,27,28, 29,30,32,33,34,35,38, 39,40		L	L	L	L	L	L				
F99		5,7,11,14,18,23,24, 27,28,32,36,38	2,3,6,8,13,15,16,17, 19,22,25,26,29,30,31, 33,34,35,39,40		H	H	H	H	H	H				
F100		5,7,11,14,15,18,23, 24,27,28,31,32,36,38	2,3,6,8,13,16,17,19, 22,25,26,29,30,33,34, 35,39,40		L	L	L	L	L	L				
F101		3,5,6,7,11,18,23,30, 34,35,36,39	2,8,13,14,15,16,17, 19,22,24,25,26,27,28, 29,31,32,33,38,40		L	L	L	L	L	L				
F102		3,5,6,7,11,15,18,23, 30,31,34,35,36,39	2,8,13,14,16,17,19, 22,24,25,26,27,28,29, 32,33,38,40		L	L	L	L	L	L				

See footnotes at end of table.

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TABLE VIII. Functional test - continued.

Test condition	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	4Y	5Y	6Y	Min	Max		
F103	Functional	3,5,6,7,11,14,18,23, 24,27,28,30,32,34,35, 36,38,39	2,8,13,15,16,17,19, 22,25,26,29,31,33,40	2/ 3/	H	H	H	H	H	H	1/			11
F104		3,5,6,7,11,14,15,18, 23,24,27,28,30,31,32, 34,35,36,38,39	2,8,13,16,17,19,22, 25,26,29,33,40		L	L	L	L	L	L				
F105		2,5,7,11,13,16,18,19, 23,26,29,36	3,6,8,14,15,17,22,24, 25,27,28,30,31,32,33, 34,35,38,39,40		L	L	L	L	L	L				
F106		2,5,7,11,13,15,16,18, 19,23,26,29,31,36	3,6,8,14,17,22,24,25, 27,28,30,32,33,34,35, 38,39,40		L	L	L	L	L	L				
F107		2,5,7,11,13,14,16,18, 19,23,24,26,27,28,29, 32,36,38	3,6,8,15,17,22,25,30, 31,33,34,35,39,40		H	H	H	H	H	H				
F108		2,5,7,11,13,14,15,16, 18,19,23,24,26,27,28, 29,31,32,36,38	3,6,8,17,22,25,30,33, 34,35,39,40		L	L	L	L	L	L				
F109		2,3,5,6,7,11,13,16, 18,19,23,26,29,30,34, 35,36,39	8,14,15,17,22,24,25, 27,28,31,32,33,38,40		L	L	L	L	L	L				
F110		2,3,5,6,7,11,13,15, 16,18,19,23,26,29,30, 31,34,35,36,39	8,14,17,22,24,25,27, 28,32,33,38,40		L	L	L	L	L	L				
F111		2,3,5,6,7,11,13,14, 16,18,19,23,24,26,27, 28,29,30,32,34,35,36, 38,39	8,15,17,22,25,31,33, 40		H	H	H	H	H	H				

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TABLE VIII. Functional test - continued.

Test condition	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	4Y	5Y	6Y	Min	Max		
F112	Functional	2,3,5,6,7,11,13,14, 15,16,18,19,23,24,26, 27,28,29,30,31,32,34, 35,36,38,39	8,17,22,25,33,40	2/ 3/	L	L	L	L	L	L		1/		11
F113		5,7,8,11,17,18,22,23, 25,33,36,40	2,3,6,13,14,15,16,19, 24,26,27,28,29,30,31, 32,34,35,38,39		L	L	L	L	L	L				
F114		5,7,8,11,15,17,18,22, 23,25,31,33,36,40	2,3,6,13,14,16,19,24, 26,27,28,29,30,32,34, 35,38,39		L	L	L	L	L	L				
F115		5,7,8,11,14,17,18,22, 23,24,25,27,28,32,33, 36,38,40	2,3,6,13,15,16,19,26, 29,30,31,34,35,39		H	H	H	H	H	H				
F116		5,7,8,11,14,15,17,18, 22,23,24,25,27,28,31, 32,33,36,38,40	2,3,6,13,16,19,26,29, 30,34,35,39		L	L	L	L	L	L				
F117		3,5,6,7,8,11,17,18, 22,23,25,30,33,34,35, 36,39,40	2,13,14,15,16,19,24, 26,27,28,29,31,32,38		L	L	L	L	L	L				
F118		3,5,6,7,8,11,15,17, 18,22,23,25,30,31,33, 34,35,36,39,40	2,13,14,16,19,24,26, 27,28,29,32,38		L	L	L	L	L	L				
F119		3,5,6,7,8,11,14,17, 18,22,23,24,25,27,28, 30,32,33,34,35,36,38, 39,40	2,13,15,16,19,26,29, 31		H	H	H	H	H	H				
F120		3,5,6,7,8,11,14,15, 17,18,22,23,24,25,27, 28,30,31,32,33,34,35, 36,38,39,40	2,13,16,19,26,29		L	L	L	L	L	L				

See footnotes at end of table.

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TABLE VIII. Functional test - continued.

Test condition	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	4Y	5Y	6Y	Min	Max		
F121	Functional	2,5,7,8,11,13,16,17, 18,19,22,23,25,26,29, 33,36,40	3,6,14,15,24,27,28, 30,31,32,34,35,38,39	2/ 3/	L	L	L	L	L	L		1/		11
F122		2,5,7,8,11,13,15,16, 17,18,19,22,23,25,26, 29,31,33,36,40	3,6,14,24,27,28,30, 32,34,35,38,39		L	L	L	L	L	L				
F123		2,5,7,8,11,13,14,16, 17,18,19,22,23,24,25, 26,27,28,29,32,33,36, 38,40	3,6,15,30,31,34,35, 39		H	H	H	H	H	H				
F124		2,5,7,8,11,13,14,15, 16,17,18,19,22,23,24, 25,26,27,28,29,31,32, 33,36,38,40	3,6,30,34,35,39		L	L	L	L	L	L				
F125		2,3,5,6,7,8,11,13,16, 17,18,19,22,23,25,26, 29,30,33,34,35,36,39, 40	14,15,24,27,28,31,32, 38		L	L	L	L	L	L				
F126		2,3,5,6,7,8,11,13,15, 16,17,18,19,22,23,25, 26,29,30,31,33,34,35, 36,39,40	14,24,27,28,32,38		L	L	L	L	L	L				
F127		2,3,5,6,7,8,11,13,14, 16,17,18,19,22,23,24, 25,26,27,28,29,30,32, 33,34,35,36,38,39,40	15,31		H	H	H	H	H	H				

See footnote at end of table.

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TABLE VIII. Functional test - continued.

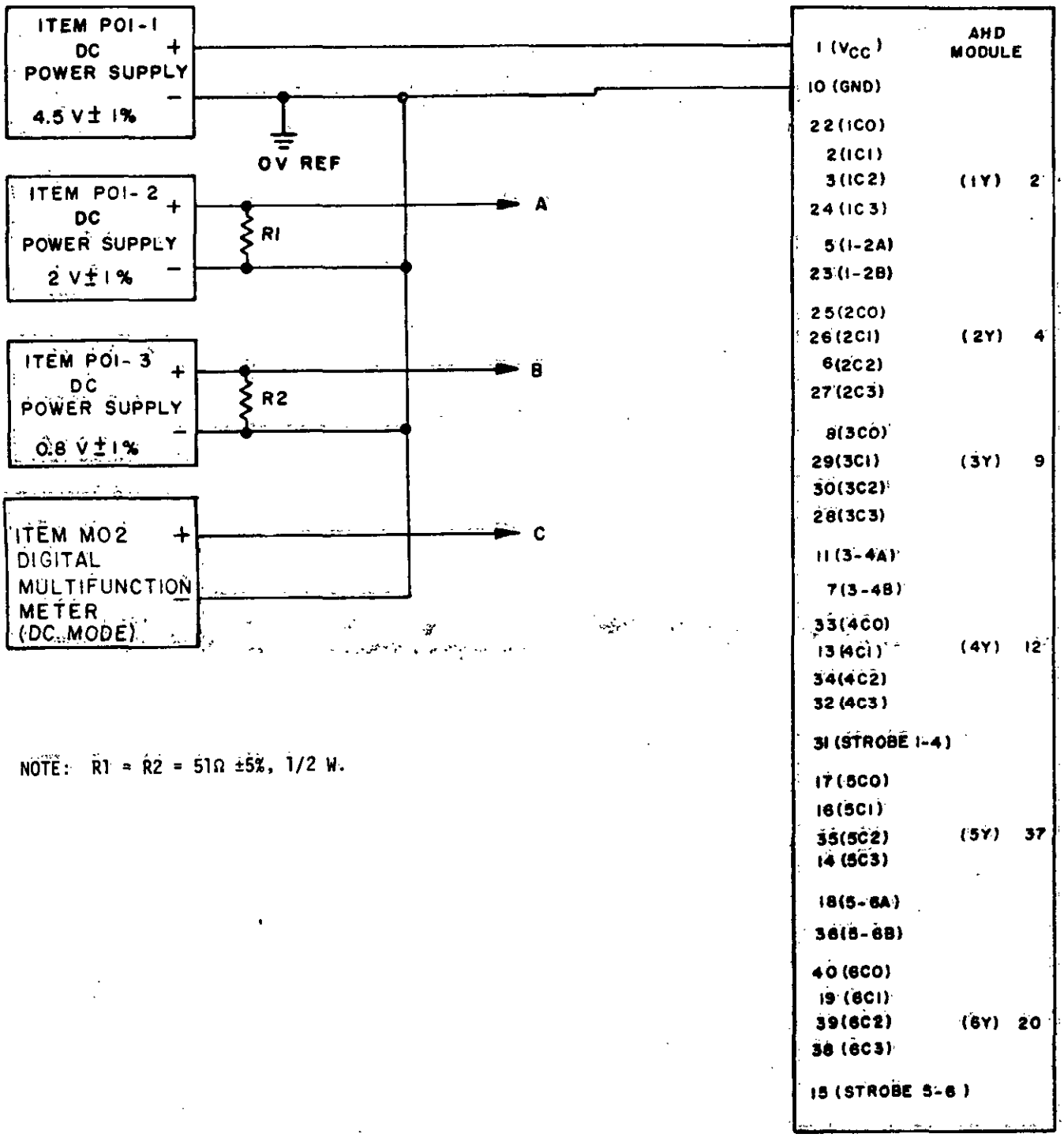
Test condition	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	4Y	5Y	6Y	Min	Max		
F128	Functional	2,3,5,6,7,8,11,13,14, 15,16,17,18,19,22,23, 24,25,26,27,28,29,30, 31,32,33,34,35,36,38, 39,40	Open	2/ 3/	L	L	L	L	L	L	1/			11

1/ Output logic states:
L = Low level = 0.0 to 0.5 V.
H = High level = 2.5 to 4.5 V.

2/ For test conditions F1-F128, verification of the output logic states in footnote 1 is obtained by connecting (item M02) point C, to each output pin in succession after input conditions have been accomplished.

3/ 1Y, 2Y, 3Y, 4Y, 5Y, 6Y output logic states are measured at output pins 21, 4, 9, 12, 37, and 20, respectively.

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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

* FIGURE 11. Functional test circuit for table VIII.

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TABLE IX. Input clamp diode voltage test.

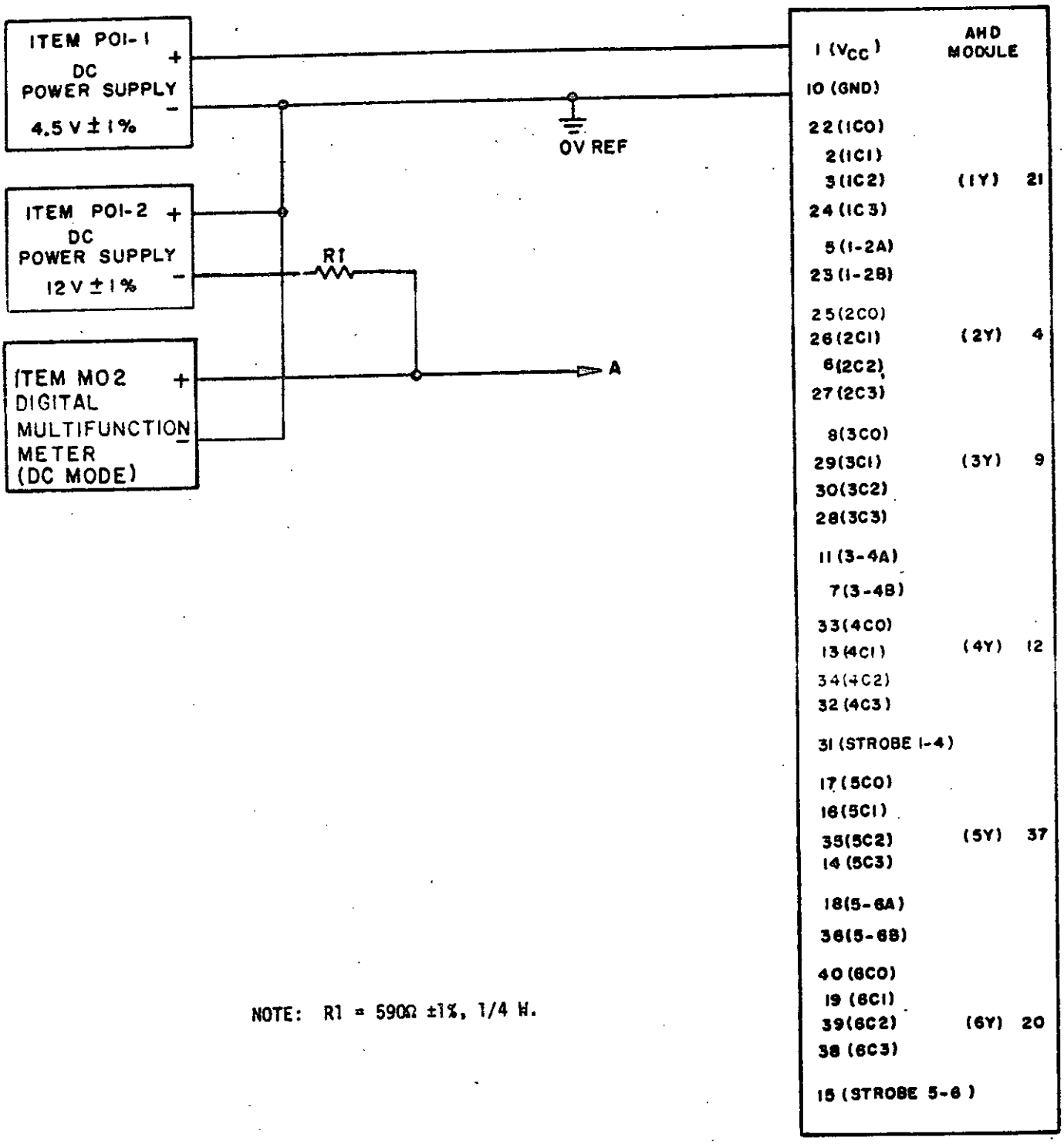
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
G1	1C0 Input clamp diode voltage	22	-0.4	-1.2	V	12
G2	1C1	2				
G3	1C2	3				
G4	1C3	24				
G5	2C0	25				
G6	2C1	26				
G7	2C2	6				
G8	2C3	27				
G9	3C0	8				
G10	3C1	29				
G11	3C2	30				
G12	3C3	28				
G13	4C0	33				
G14	4C1	13				
G15	4C2	34				
G16	4C3	32				
G17	5C0	17				
G18	5C1	16				
G19	5C2	35				

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TABLE IX. Input clamp diode voltage test - continued.

Test condition	Parameter	Connect A to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
G20	5C3 Input clamp diode voltage	14	-0.4	-1.2	V	12
G21	6C0	40				
G22	6C1	19				
G23	6C2	39				
G24	6C3	38				
G25	1-2A	5				
G26	1-2B	23				
G27	3-4A	11				
G28	3-4B	7				
G29	5-6A	18				
G30	5-6B	36				
G31	STROBE 1-4	31				
G32	STROBE 5-6	15				

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NOTE: R1 = 590Ω ±1%, 1/4 W.

* FIGURE 12. Input clamp diode voltage test circuit for table IX.

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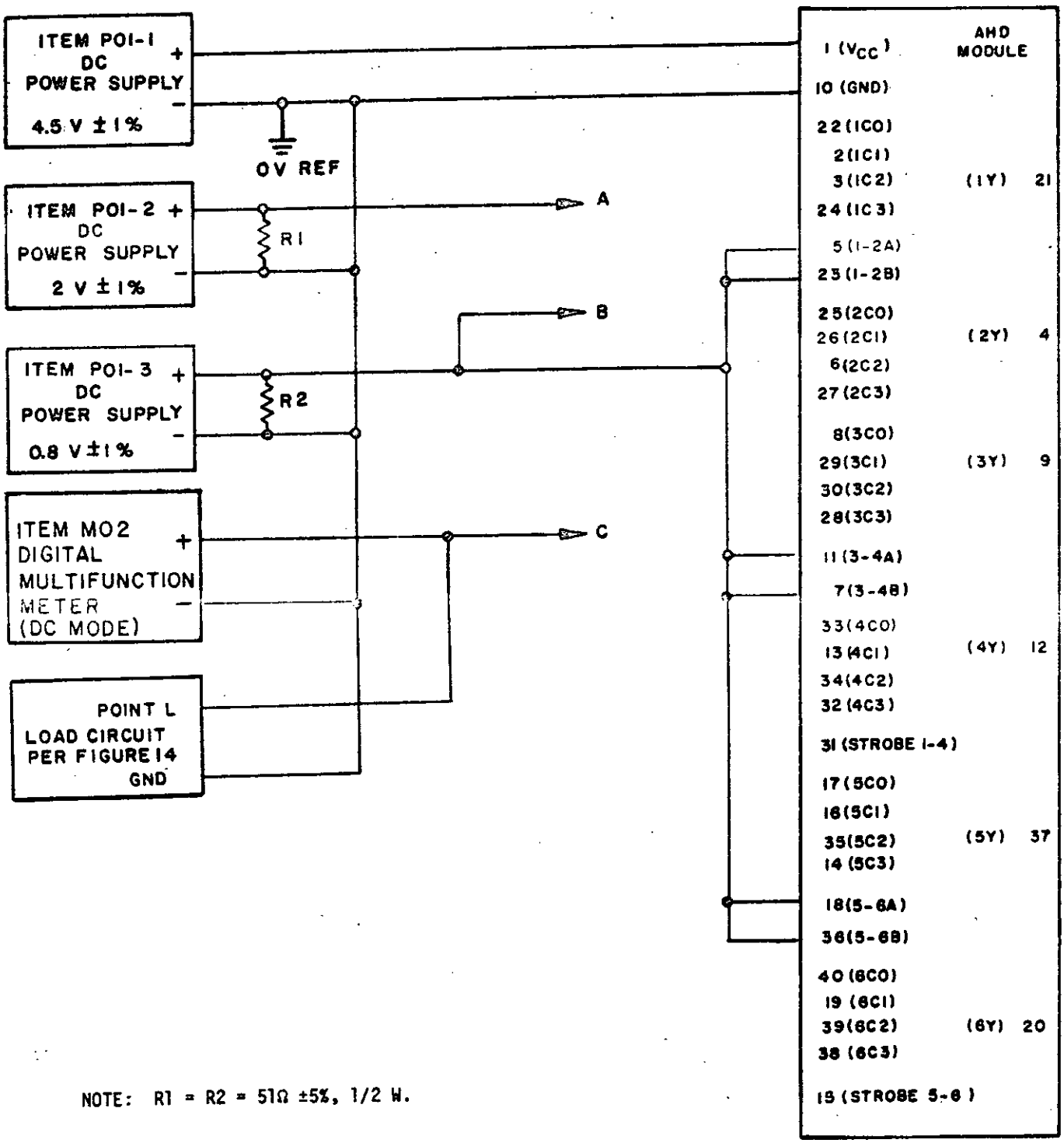
TABLE X. Output voltage (low-level) test.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H1	1Y	Output voltage, low level	15,31	Open	21	0	500	mV	13
H2	2Y				4				
H3	3Y				9				
H4	4Y				12				
H5	5Y				37				
H6	6Y				20				

TABLE XI. Output voltage (high-level) test.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
J1	1Y	Output voltage, high level	22	8,15,17, 25,31,33, 40	21	2.5	4.5	V	13
J2	2Y		25	8,15,17, 22,31,33, 40	4				
J3	3Y		8	15,17,22, 25,31,33, 40	9				
J4	4Y		33	8,15,17, 22,25,31, 40	12				
J5	5Y		17	8,15,22, 25,31,33, 40	37				
J6	6Y		40	8,15,17, 22,25,31, 33	20				

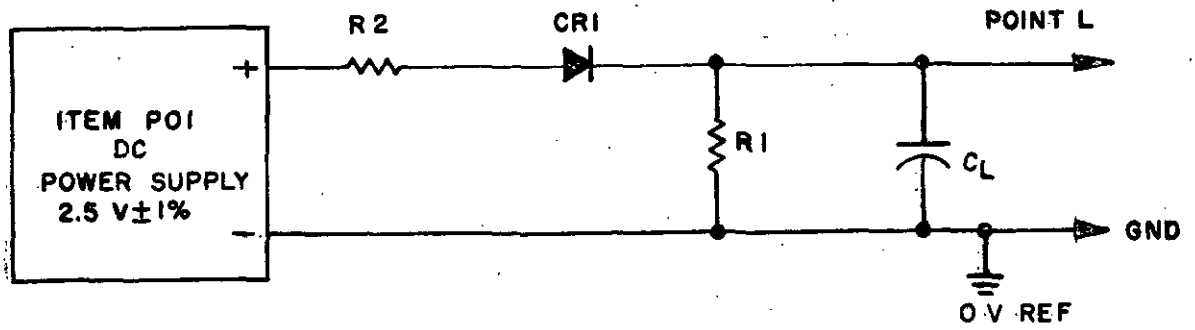
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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

* FIGURE 13. Output voltage (low-level and high-level) test circuit for tables X and XI.

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NOTES:
1. CRL = 1N3600, 1N4150, or equivalent.
2. R2 = Select value for a current flow of 20.0 mA ±5% out of point L with point L held at 0.5 V.
3. R1 = 2.49 kΩ ±1%, 1/8 W.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions N1-N6, P1-P6, Q1-Q42, R1-R42, S1-S6, and T1-T6.
5. C_L = 150 ±5 pF (including probe and parasitic capacitance) for test conditions N7-N12, P7-P12, Q43-Q84, and R43-R84.
6. Capacitor C_L is not required for dc electrical test conditions H1-H6 and J1-J6.
7. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm).

* FIGURE 14. AHD logic load circuit.

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TABLE XII. Input current (high-level test).

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	1C0 Input current, high level	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,23,24,25,26, 27,28,29,30,32,33,34,35, 36,38,39,40	22	0	50	μA	15
K2	1C1	3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33, 34,35,36,38,39,40	2				
K3	1C2	2,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33, 34,35,36,38,39,40	3				
K4	1C3	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,25,26, 27,28,29,30,31,32,33,34, 35,36,38,39,40	24				
K5	2C0	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,26, 27,28,29,30,32,33,34,35, 36,38,39,40	25				
K6	2C1	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 27,28,29,30,31,32,33,34, 35,36,38,39,40	26				
K7	2C2	2,3 5,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33, 34,35,36,38,39,40	6				

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TABLE XII. Input current (high-level test) - continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
K8	2C3 Input current, high level	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,28,29,30,31,32,33,34, 35,36,38,39,40	27	0	50	µA	15
K9	3C0	2,3,5,6,7,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,32,33,34, 35,36,38,39,40	8				
K10	3C1	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,30,31,32,33,34, 35,36,38,39,40	29				
K11	3C2	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,31,32,33,34, 35,36,38,39,40	30				
K12	3C3	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,29,30,31,32,33,34, 35,36,38,39,40	28				
K13	4C0	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,32,34,35, 36,38,39,40	33				
K14	4C1	2,3,5,6,7,8,11,14,15,16, 17,18,19,22,23,24,25,26, 27,28,29,30,31,32,33,34, 35,36,38,39,40	13				

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TABLE XII. Input current (high-level test) - continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K15	4C2 Input current, high level	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33, 35,36,38,39,40	34	0	50	µA	15
K16	4C3	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,33,34, 35,36,38,39,40	32				
K17	5C0	2,3,5,6,7,8,11,13,14,16, 18,19,22,23,24,25,26,27, 28,29,30,31,32,33,34,35, 36,38,39,40	17				
K18	5C1	2,3,5,6,7,8,11,13,14,15, 17,18,19,22,23,24,25,26, 27,28,29,30,31,32,33,34, 35,36,38,39,40	16				
K19	5C2	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33, 34,36,38,39,40	35				
K20	5C3	2,3,5,6,7,8,11,13,15,16, 17,18,19,22,23,24,25,26, 27,28,29,30,31,32,33,34, 35,36,38,39,40	14				
K21	6C0	2,3,5,6,7,8,11,13,14,16, 17,18,19,22,23,24,25,26, 27,28,29,30,31,32,33,34, 35,36,38,39	40				

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TABLE XII. Input current (high-level test) - continued.

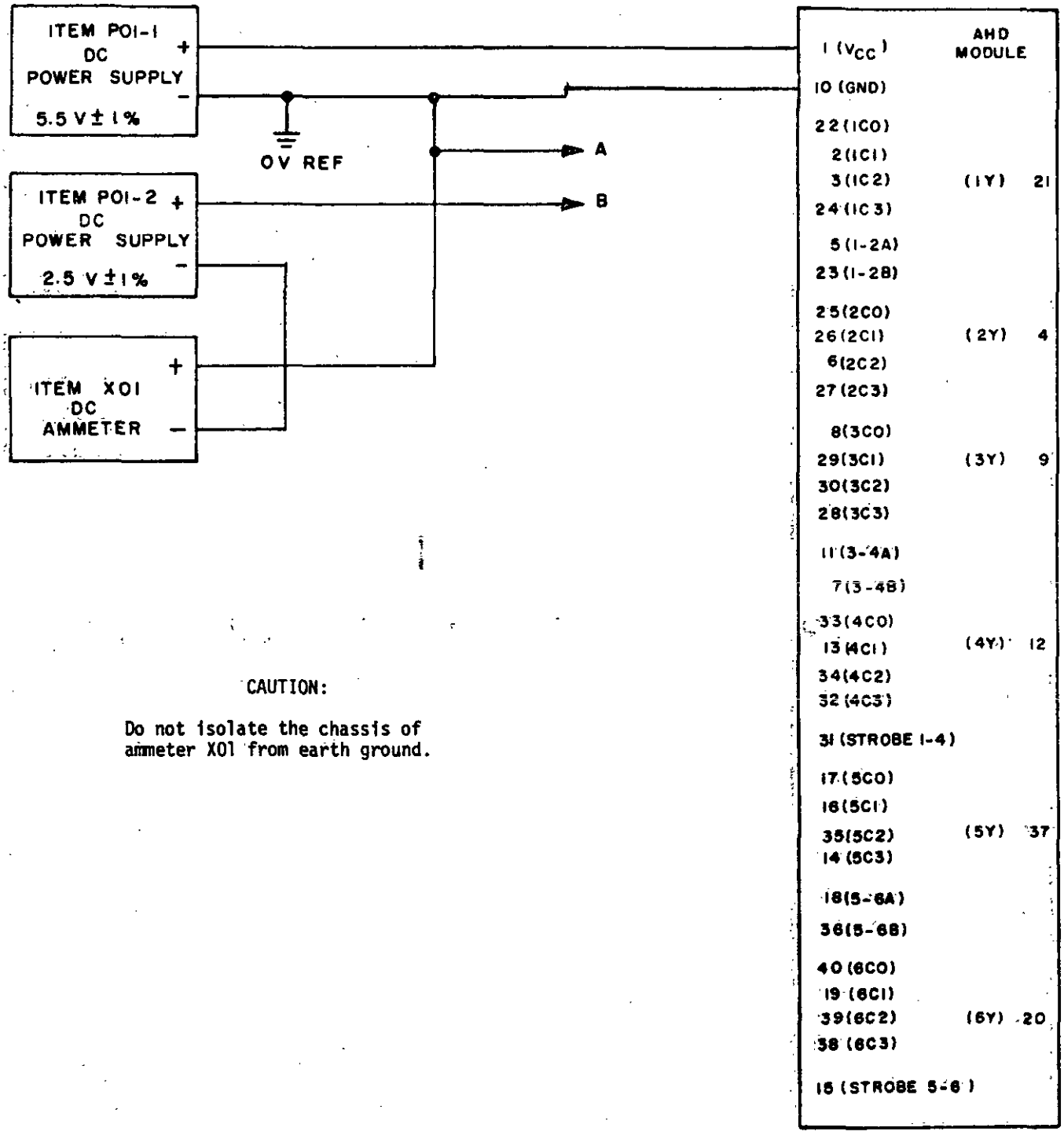
Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
K22	6C1 Input current, high level	2,3,5,6,7,8,11,13,14,15, 16,17,18,22,23,24,25,26, 27,28,29,30,31,32,33,34, 35,36,38,39,40	19	0	50	µA	15
K23	6C2	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33, 34,35,36,38,40	39				
K24	6C3	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33, 34,35,36,39,40	38				
K25	1-2A	2,3,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33, 34,35,36,38,39,40	5				
K26	1-2B	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,24,25,26, 27,28,29,30,31,32,33,34, 35,36,38,39,40	23				
K27	3-4A	2,3,5,6,7,8,13,14,15,16, 17,18,19,22,23,24,25,26, 27,28,29,30,31,32,33,34, 35,36,38,39,40	11				
K28	3-4B	2,3,5,6,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33, 34,35,36,38,39,40	7				

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TABLE XII. Input current (high-level test) - continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
K29	5-6A Input current, high level	2,3,5,6,7,8,11,13,14,15, 16,17,19,22,23,24,25,26, 27,28,29,30,31,32,33,34, 35,36,38,39,40	18	0	50	µA	15
K30	5-6B	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,31,32,33, 34,35,38,39,40	36		50		
K31	STROBE 1-4	2,3,5,6,7,8,11,13,14,15, 16,17,18,19,22,23,24,25, 26,27,28,29,30,32,33,34, 35,36,38,39,40	31		200		
K32	STROBE 5-6	2,3,5,6,7,8,11,13,14,16, 17,18,19,22,23,24,25,26, 27,28,29,30,31,32,33,34, 35,36,38,39,40	15		100		

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 15. Input current (high-level) test circuit for table XII.

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TABLE XIII. Input current (low-level) test.

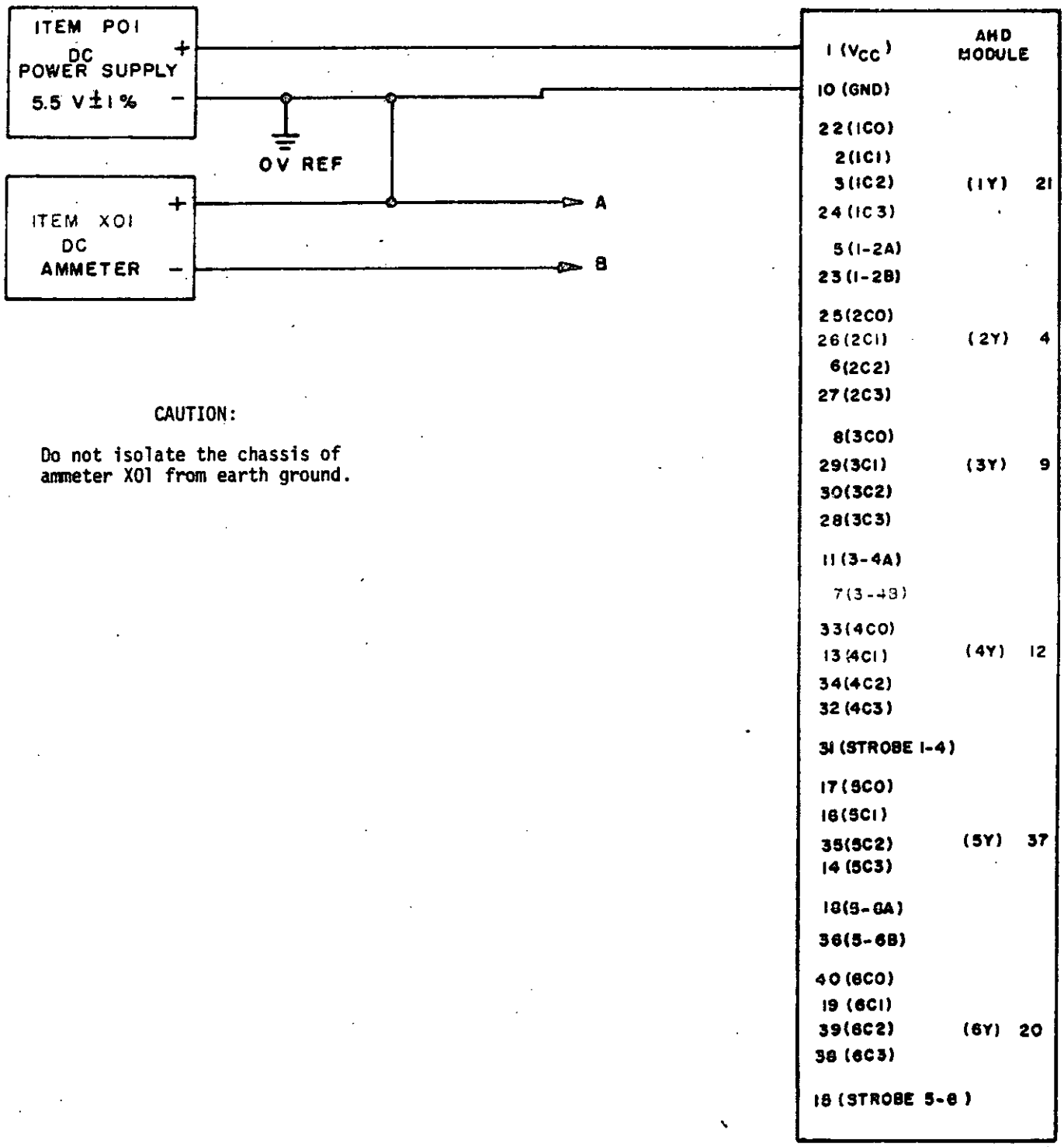
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	1C0 Input current, low level	5,23,31	22	-0.1	-2.2	mA	16
L2	1C1	23,31	2				
L3	1C2	5,31	3				
L4	1C3	31	24				
L5	2C0	5,23,31	25				
L6	2C1	23,31	26				
L7	2C2	5,31	6				
L8	2C3	31	27				
L9	3C0	7,11,31	8				
L10	3C1	7,31	29				
L11	3C2	11,31	30				
L12	3C3	31	28				
L13	4C0	7,11,31	33				
L14	4C1	7,31	13				
L15	4C2	11,31	34				
L16	4C3	31	32				
L17	5C0	15,18,36	17				
L18	5C1	15,36	16				
L19	5C2	15,18	35				

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TABLE XIII. Input current (low-level) test - continued.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L20	5C3	Input current, low level	15	14	-0.1	-2.2	mA	16
L21	6C0		15,18,36	40				
L22	6C1		15,36	19				
L23	6C2		15,18	39				
L24	6C3		15	38				
L25	1-2A		Open	5				
L26	1-2B			23				
L27	3-4A			11				
L28	3-4B			7				
L29	5-6A			18				
L30	5-6B			36				
L31	STROBE 1-4			31		-8.8		
L32	STROBE 5-6			15		-4.4		

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 16. Input current (low-level) test circuit for table XIII.

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TABLE XIV. Power supply current (low-level outputs) test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
M	Power supply current, low-level outputs	45	210	mA	17

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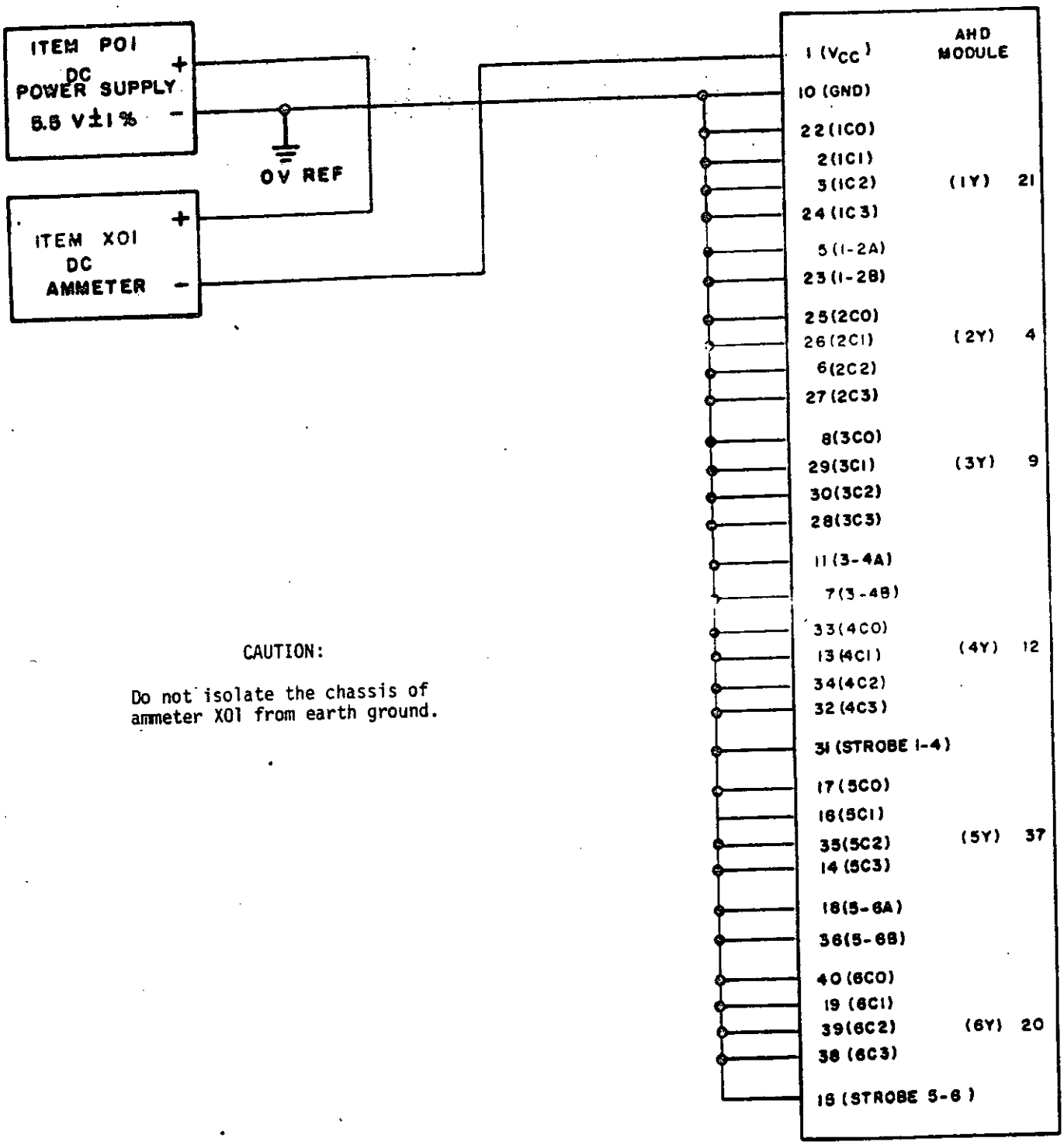


FIGURE 17. Power supply current (low-level outputs) test circuit for table XIV.

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TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	1Y	Transition time (t_{TLH}), C_L = 50 pF	Open	5,23,31	22	21	2	15	ns	18,19
N2	2Y			5,23,31	25	4				
N3	3Y			7,11,31	8	9				
N4	4Y			7,11,31	33	12				
N5	5Y			15,18,36	17	37				
N6	6Y			15,18,36	40	20				
N7	1Y	Transition time (t_{TLH}), C_L = 150 pF		5,23,31	22	21		27		
N8	2Y			5,23,31	25	4				
N9	3Y			7,11,31	8	9				
N10	4Y			7,11,31	33	12				
N11	5Y			15,18,36	17	37				
N12	6Y			15,18,36	40	20				
P1	1Y	Transition time (t_{THL}), C_L = 50 pF		5,23,31	22	21		6		
P2	2Y			5,23,31	25	4				
P3	3Y			7,11,31	8	9				
P4	4Y			7,11,31	33	12				

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TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P5	5Y	Transition time (t_{THL}), C_L = 50 pF	Open	15,18,36	17	37	2	6	ns	18,19
P6	6Y	Transition time (t_{THL}), C_L = 50 pF		15,18,36	40	20		6		
P7	1Y	Transition time (t_{THL}), C_L = 150 pF		5,23,31	22	21		11		
P8	2Y			5,23,31	25	4				
P9	3Y			7,11,31	8	9				
P10	4Y			7,11,31	33	12				
P11	5Y			15,18,36	17	37				
P12	6Y			15,18,36	40	20				
Q1	1C0 to 1Y	Propagation delay time (t_{PLH}), C_L = 50 pF		5,23,31	22	21				18,20
Q2	1C1 to 1Y		5	23,31	2					
Q3	1C2 to 1Y		23	5,31	3					
Q4	1C3 to 1Y		5,23	31	24					
Q5	2C0 to 2Y		Open	5,23,31	25	4				
Q6	2C1 to 2Y		5	23,31	26	4				
Q7	2C2 to 2Y		23	5,31	6	4				

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TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q8	2C3 to 2Y Propagation delay time (t_{PLH}), C_L = 50 pF	5,23	31	27	4	2	11	ns	18,20
Q9	3C0 to 3Y	Open	7,11,31	8	9				
Q10	3C1 to 3Y	11	7,31	29					
Q11	3C2 to 3Y	7	11,31	30					
Q12	3C3 to 3Y	7,11	31	28					
Q13	4C0 to 4Y	Open	7,11,31	33	12				
Q14	4C1 to 4Y	11	7,31	13					
Q15	4C2 to 4Y	7	11,31	34					
Q16	4C3 to 4Y	7,11	31	32					
Q17	5C0 to 5Y	Open	15,18,36	17	37				
Q18	5C1 to 5Y	18	15,36	16					
Q19	5C2 to 5Y	36	15,18	35					
Q20	5C3 to 5Y	18,36	15	14					
Q21	6C0 to 6Y	Open	15,18,36	40	20				
Q22	6C1 to 6Y	18	15,36	19					
Q23	6C2 to 6Y	36	15,18	39					
Q24	6C3 to 6Y	18,36	15	38					

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TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q25	1-2A to 1Y Propagation delay time (t_{PLH}), C_L = 50 pF	2	22,23,31	5	21	2	20	ns	18,20
Q26	1-2B to 1Y	3	5,22,31	23	21				
Q27	1-2A to 2Y	26	23,25,31	5	4				
Q28	1-2B to 2Y	6	5,25,31	23	4				
Q29	3-4A to 3Y	29	7,8,31	11	9				
Q30	3-4B to 3Y	30	8,11,31	7	9				
Q31	3-4A to 4Y	13	7,31,33	11	12				
Q32	3-4B to 4Y	34	11,31,33	7	12				
Q33	5-6A to 5Y	16	15,17,36	18	37				
Q34	5-6B to 5Y	35	15,17,18	36	37				
Q35	5-6A to 6Y	19	15,36,40	18	20				
Q36	5-6B to 6Y	39	15,18,40	36	20				
Q37	STROBE 1-4 to 1Y	22	5,23	31	21	2.0	17.5		18,21
Q38	STROBE 1-4 to 2Y	25	5,23	31	4				
Q39	STROBE 1-4 to 3Y	8	7,11	31	9				
Q40	STROBE 1-4 to 4Y	33	7,11	31	12				

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TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q41	STROBE 5-6 to 5Y Propagation delay time (t_{PLH}), C_L = 50 pF	17	18,36	15	37	2.0	17.5	ns	18,21
Q42	STROBE 5-6 to 6Y Propagation delay time (t_{PLH}), C_L = 50 pF	40	18,36	15	20	2.0	17.5		18,21
Q43	1C0 to 1Y Propagation delay time (t_{PLH}), C_L = 150 pF	Open	5,23,31	22	21	4.0	14.5		18,20
Q44	1C1 to 1Y	5	23,31	2					
Q45	1C2 to 1Y	23	5,31	3					
Q46	1C3 to 1Y	5,23	31	24					
Q47	2C0 to 2Y	Open	5,23,31	25	4				
Q48	2C1 to 2Y	5	23,31	26					
Q49	2C2 to 2Y	23	5,31	6					
Q50	2C3 to 2Y	5,23	31	27					
Q51	3C0 to 3Y	Open	7,11,31	8	9				
Q52	3C1 to 3Y	11	7,31	29					
Q53	3C2 to 3Y	7	11,31	30					
Q54	3C3 to 3Y	7,11	31	28					

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TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q55	4C0 to 4Y Propagation delay time (t_{PLH}), C_L = 150 pF	Open	7,11,31	33	12	4.0	14.5	ns	18,20
Q56	4C1 to 4Y	11	7,31	13					
Q57	4C2 to 4Y	7	11,31	34					
Q58	4C3 to 4Y	7,11	31	32					
Q59	5C0 to 5Y	Open	15,18,36	17	37				
Q60	5C1 to 5Y	18	15,36	16					
Q61	5C2 to 5Y	36	15,18	35					
Q62	5C3 to 5Y	18,36	15	14					
Q63	6C0 to 6Y	Open	15,18,36	40	20				
Q64	6C1 to 6Y	18	15,36	19					
Q65	6C2 to 6Y	36	15,18	39					
Q66	6C3 to 6Y	18,36	15	38					
Q67	1-2A to 1Y	2	22,23,31	5	21		23.5		
Q68	1-2B to 1Y	3	5,22,31	23	21				
Q69	1-2A to 2Y	26	23,25,31	5	4				
Q70	1-2B to 2Y	6	5,25,31	23	4				
Q71	3-4A to 3Y	29	7,8,31	11	9				

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TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q72	3-4B to 3Y	Propagation delay time (t_{PLH}), C_L = 150 pF	30	8,11,31	7	9	4.0	23.5	ns	18,20
Q73	3-4A to 4Y		13	7,31,33	11	12				
Q74	3-4B to 4Y		34	11,31,33	7	12				
Q75	5-6A to 5Y		16	15,17,36	18	37				
Q76	5-6B to 5Y		35	15,17,18	36	37				
Q77	5-6A to 6Y		19	15,36,40	18	20				
Q78	5-6B to 6Y		39	15,18,40	36	20				
Q79	STROBE 1-4 to 1Y		22	5,23	31	21	4	20		18,21
Q80	STROBE 1-4 to 2Y		25	5,23	31	4				
Q81	STROBE 1-4 to 3Y		8	7,11	31	9				
Q82	STROBE 1-4 to 4Y		33	7,11	31	12				
Q83	STROBE 5-6 to 5Y		17	18,36	15	37				
Q84	STROBE 5-6 to 6Y		40	18,36	15	20				
R1	1C0 to 1Y	Propagation delay time (t_{PHL}), C_L = 50 pF	Open	5,23,31	22	21	2.0	10.5		18,20

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TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R2	1C1 to 1Y Propagation delay time (t_{PHL}), C_L = 50 pF	5	23,31	2	21	2.0	10.5	ns	18,20
R3	1C2 to 1Y	23	5,31	3	21				
R4	1C3 to 1Y	5,23	31	24	21				
R5	2C0 to 2Y	Open	5,23,31	25	4				
R6	2C1 to 2Y	5	23,31	26					
R7	2C2 to 2Y	23	5,31	6					
R8	2C3 to 2Y	5,23	31	27					
R9	3C0 to 3Y	Open	7,11,31	8	9				
R10	3C1 to 3Y	11	7,31	29					
R11	3C2 to 3Y	7	11,31	30					
R12	3C3 to 3Y	7,11	31	28					
R13	4C0 to 4Y	Open	7,11,31	33	12				
R14	4C1 to 4Y	11	7,31	13					
R15	4C2 to 4Y	7	11,31	34					
R16	4C3 to 4Y	7,11	31	32					
R17	5C0 to 5Y	Open	15,18,36	17	37				
R18	5C1 to 5Y	18	15,36	16	37				

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TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R19	5C2 to 5Y Propagation delay time (t_{PHL}), C_L = 50 pF	36	15,18	35	37	2.0	10.5	ns	18,20
R20	5C3 to 5Y	18,36	15	14	37				
R21	6C0 to 6Y	Open	15,18,36	40	20				
R22	6C1 to 6Y	18	15,36	19					
R23	6C2 to 6Y	36	15,18	39					
R24	6C3 to 6Y	18,36	15	38					
R25	1-2A to 1Y	2	22,23,31	5	21		19.5		
R26	1-2B to 1Y	3	5,22,31	23	21				
R27	1-2A to 2Y	26	23,25,31	5	4				
R28	1-2B to 2Y	6	5,25,31	23	4				
R29	3-4A to 3Y	29	7,8,31	11	9				
R30	3-4B to 3Y	30	8,11,31	7	9				
R31	3-4A to 4Y	13	7,31,33	11	12				
R32	3-4B to 4Y	34	11,31,33	7	12				
R33	5-6A to 5Y	16	15,17,36	18	37				
R34	5-6B to 5Y	35	15,17,18	36	37				
R35	5-6A to 6Y	19	15,36,40	18	20				

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TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R36	5-6B to 6Y Propagation delay time (t_{PHL}), C_L = 50 pF	39	15,18,40	36	20	2.0	19.5	ns	18,20
R37	STROBE 1-4 to 1Y	22	5,23	31	21	2	15		18,21
R38	STROBE 1-4 to 2Y	25	5,23		4				
R39	STROBE 1-4 to 3Y	8	7,11		9				
R40	STROBE 1-4 to 4Y	33	7,11		12				
R41	STROBE 5-6 to 5Y	17	18,36	15	37				
R42	STROBE 5-6 to 6Y	40	18,36	15	20				
R43	1C0 to 1Y Propagation delay time (t_{PHL}), C_L = 150 pF	Open	5,23,31	22	21	4.0	13.5		18,20
R44	1C1 to 1Y	5	23,31	2					
R45	1C2 to 1Y	23	5,31	3					
R46	1C3 to 1Y	5,23	31	24					
R47	2C0 to 2Y	Open	5,23,31	25	4				

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TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R48	2C1 to 2Y Propagation delay time (t_{PHL}), C_L = 150 pF	5	23,31	26	4	4.0	13.5	ns	18,20
R49	2C2 to 2Y	23	5,31	6	4				
R50	2C3 to 2Y	5,23	31	27	4				
R51	3C0 to 3Y	Open	7,11,31	8	9				
R52	3C1 to 3Y	11	7,31	29					
R53	3C2 to 3Y	7	11,31	30					
R54	3C3 to 3Y	7,11	31	28					
R55	4C0 to 4Y	Open	7,11,31	33	12				
R56	4C1 to 4Y	11	7,31	13					
R57	4C2 to 4Y	7	11,31	34					
R58	4C3 to 4Y	7,11	31	32					
R59	5C0 to 5Y	Open	15,18,36	17	37				
R60	5C1 to 5Y	18	15,36	16					
R61	5C2 to 5Y	36	15,18	35					
R62	5C3 to 5Y	18,36	15	14					
R63	6C0 to 6Y	Open	15,18,36	40	20				
R64	6C1 to 6Y	18	15,36	19	20				

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TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - continued.

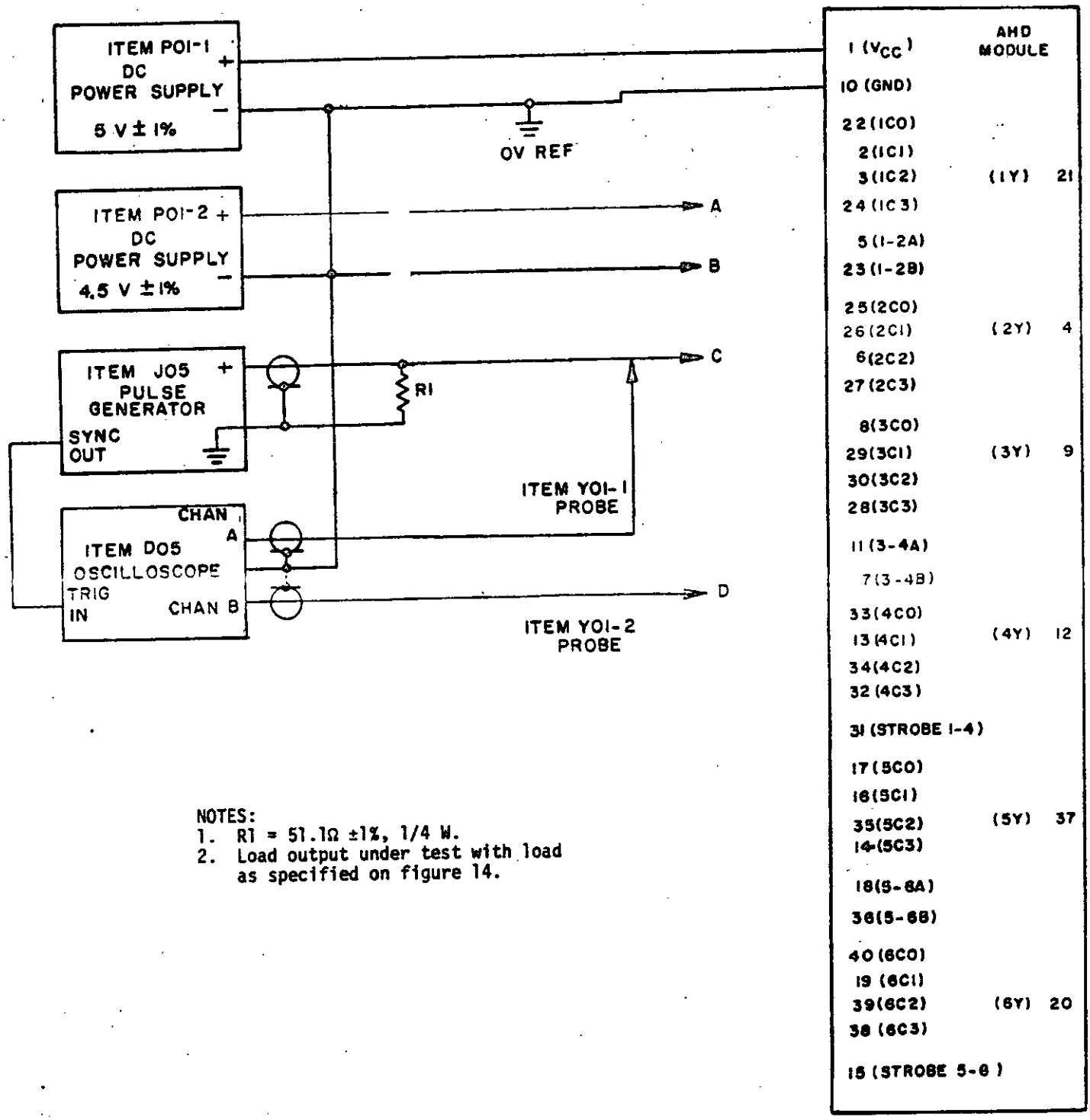
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R65	6C2 to 6Y Propagation delay time (t_{PHL}), C_L = 150 pF	36	15,18	39	20	4.0	13.5	ns	18,20
R66	6C3 to 6Y	18,36	15	38	20	4.0	13.5		
R67	1-2A to 1Y	2	22,23,31	5	21	4	23		
R68	1-2B to 1Y	3	5,22,31	23	21				
R69	1-2A to 2Y	26	23,25,31	5	4				
R70	1-2B to 2Y	6	5,25,31	23	4				
R71	3-4A to 3Y	29	7,8,31	11	9				
R72	3-4B to 3Y	30	8,11,31	7	9				
R73	3-4A to 4Y	13	7,31,33	11	12				
R74	3-4B to 4Y	34	11,31,33	7	12				
R75	5-6A to 5Y	16	15,17,36	18	37				
R76	5-6B to 5Y	35	15,17,18	36	37				
R77	5-6A to 6Y	19	15,36,40	18	20				
R78	5-6B to 6Y	39	15,18,40	36	20				
R79	STROBE 1-4 to 1Y	22	5,23	31	21		18		18,21
R80	STROBE 1-4 to 2Y	25	5,23	31	4		18		18,21

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TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R81	STROBE 1-4 Propagation delay time (t_{PHL}), C_L = 150 pF	8	7,11	31	9	4	18	ns	18,21
R82	STROBE 1-4 to 4Y	33	7,11	31	12	↓	↓	↓	↓
R83	STROBE 5-6 to 5Y	17	18,36	15	37				
R84	STROBE 5-6 to 6Y	40	18,36	15	20				

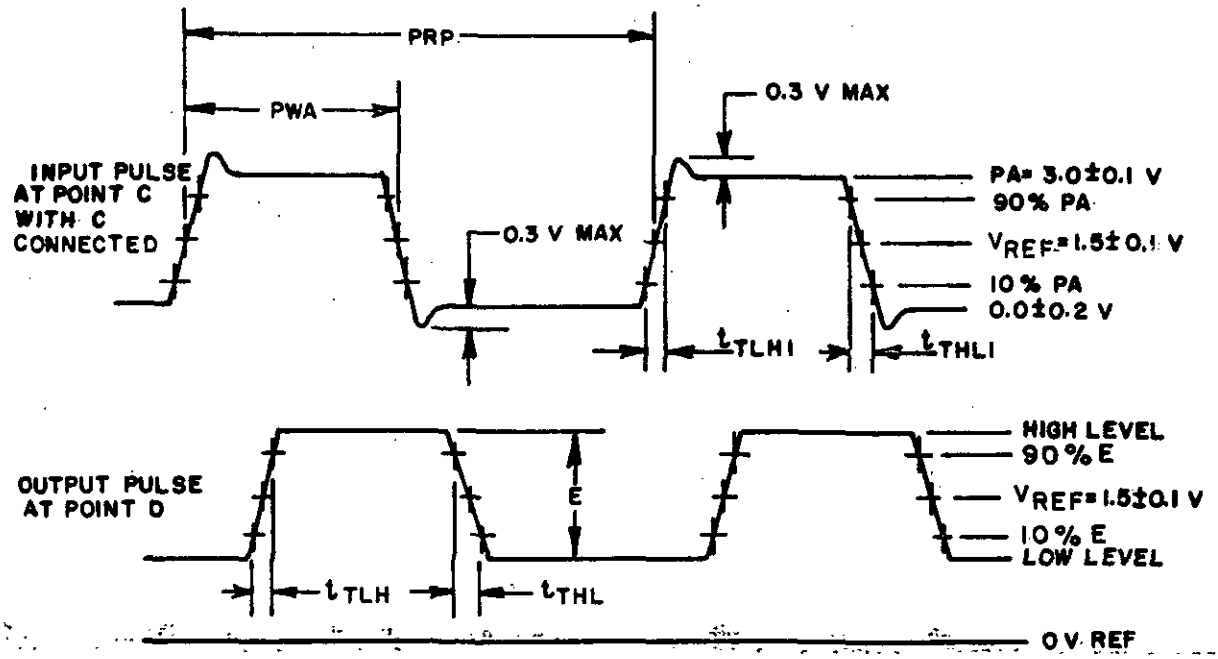
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load output under test with load as specified on figure 14.

* FIGURE 18. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XV.

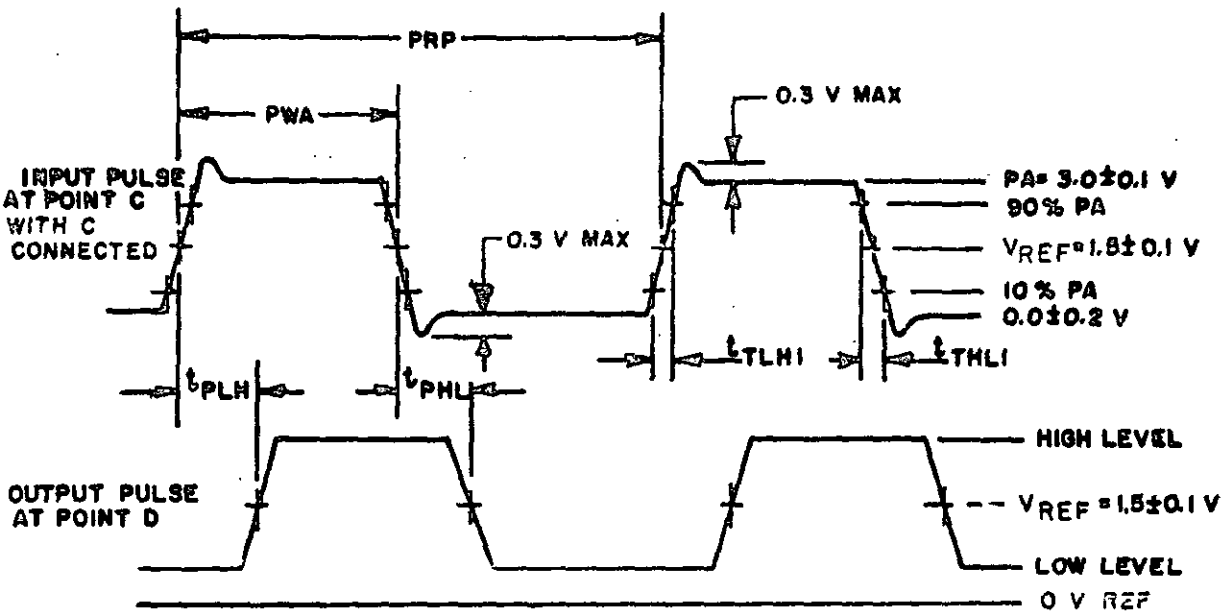
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NOTES:
1. Input pulse parameters:
   $t_{TLHI} = t_{THLI} = 4 \pm 1$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between terminator resistor R1 and input pins; porbes YO1-1, YO1-2 to output pins; and between load circuits and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

* FIGURE 19. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for table XV.

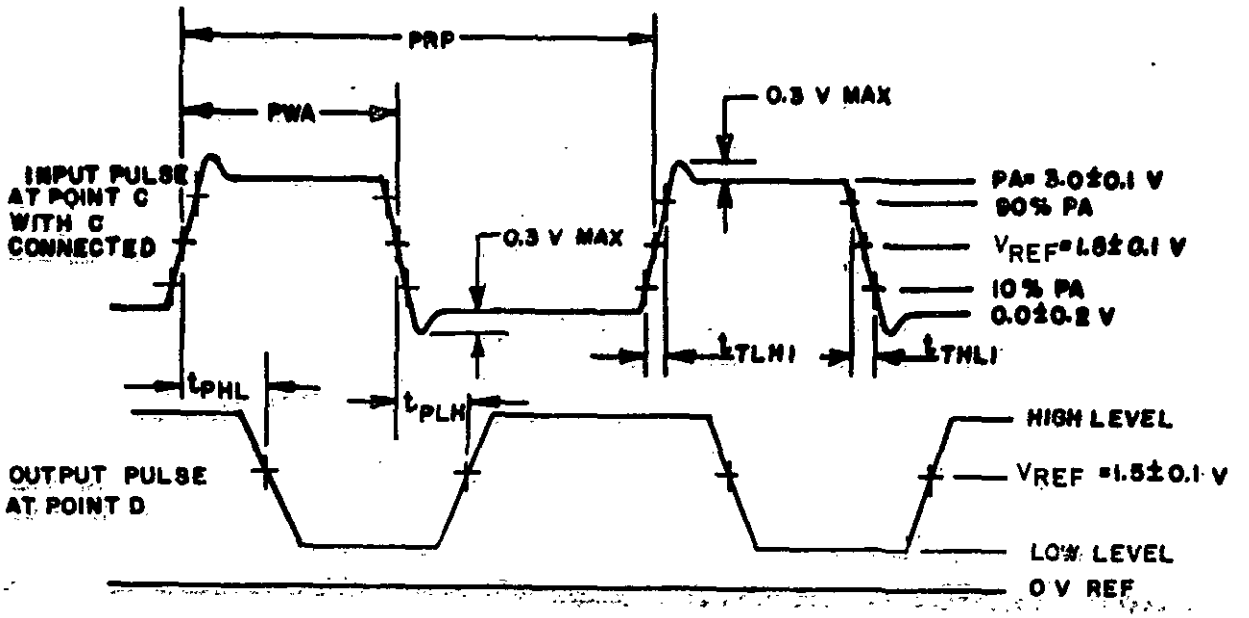
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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 4 \pm 1$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical condictors between terminator resistor R1 and input pins; probes YO1-1, YO1-2 to output pins; and between load circuits and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

* FIGURE 20. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XV.

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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 4 \pm 1$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between terminator resistor R1 and input pins; probes YO1-1, YO1-2 to output pins; and between load circuits and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

* FIGURE 21. Propagation delay time (t_{PLH} and t_{PHL}) test circuit-waveforms for table XV.

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TABLE XVI. Output crosstalk (low-level and high-level test.

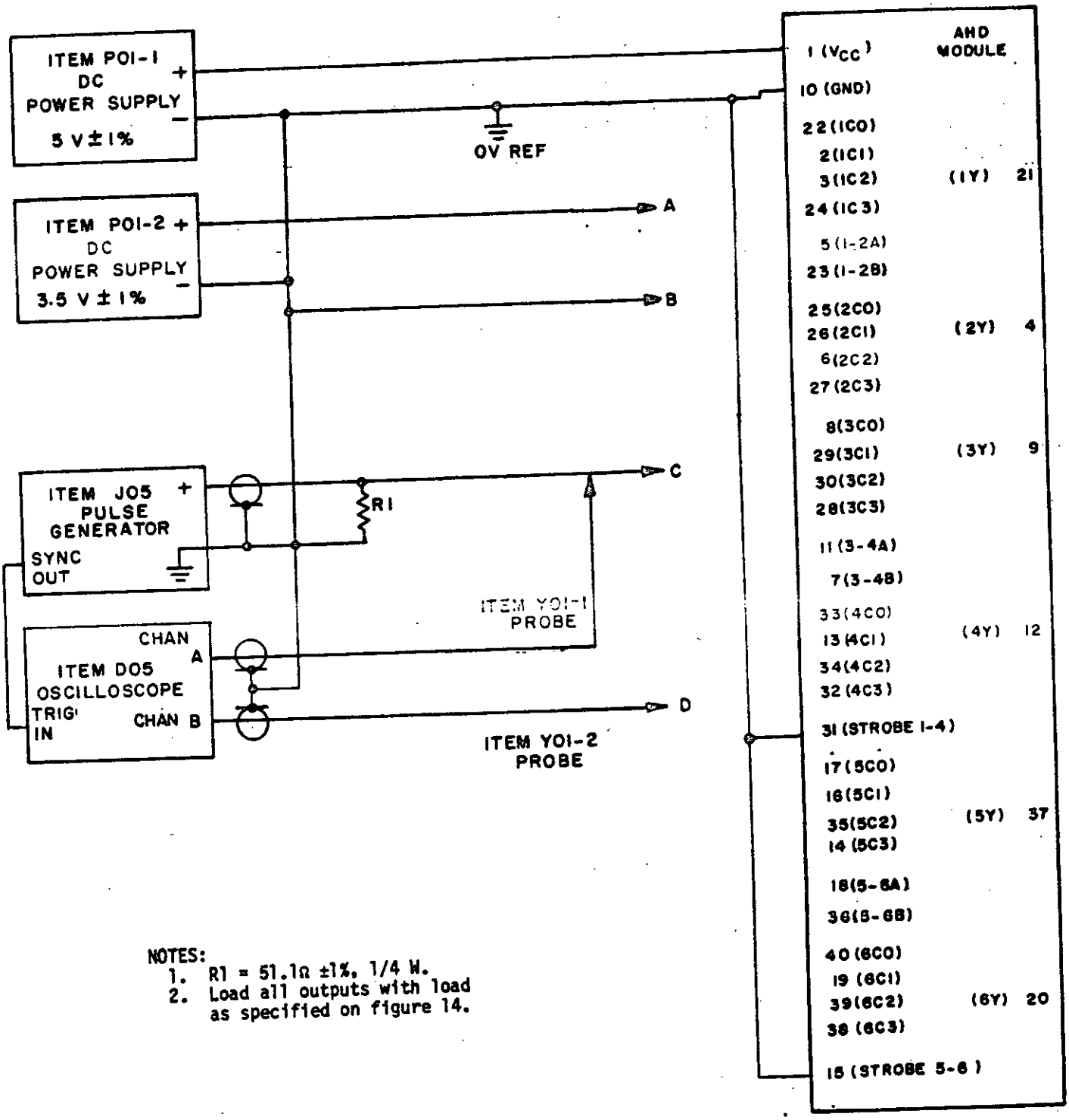
Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1Y	Output crosstalk, low level	2,3,5, 22,23	24	14,27, 28,32, 38	21	0	650	mV	22,23
S2	2Y		5,6, 23,25, 26	27	14,24, 28,32, 38	4				
S3	3Y		7,8, 11,29, 30	28	14,24, 27,32, 38	9				
S4	4Y		7,11, 13,33, 34	32	14,24, 27,28, 38	12				
S5	5Y		16,17, 18,35, 36	14	24,27, 28,32, 38	37				
S6	6Y		18,19, 36,39, 40	38	14,24, 27,28, 32	20				
T1	1Y	Output crosstalk, high level	5,23, 24	2,3,22	14,27, 28,32, 38	21		400		
T2	2Y	Output crosstalk, high level	5,23, 27	6,25, 26	14,24, 28,32, 38	4		400		
T3	3Y	Output crosstalk, high level	7,11, 28	8,29, 30	14,24, 27,32, 38	9		400		

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TABLE XVI. Output crosstalk (low-level and high-level test - continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
T4	4Y	Output crosstalk, high level	7,11, 32	13,33, 34	14,24, 27,28, 38	12	0	400	mV	22,23
T5	5Y	Output crosstalk, high level	14,18, 36	16,17, 35	24,27, 28,32, 38	37	0	400	mV	22,23
T6	6Y	Output crosstalk, high level	18,36, 38	19,39, 40	14,24, 27,28, 32	20	0	400	mV	22,23

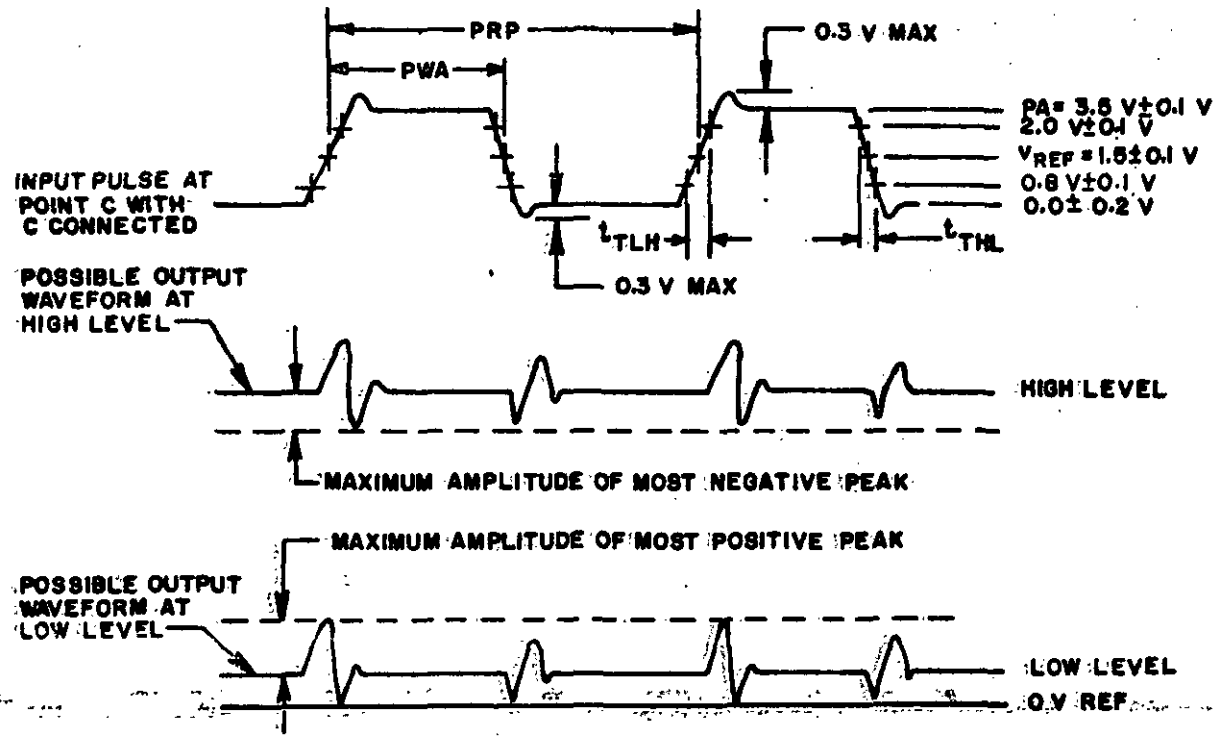
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load all outputs with load as specified on figure 14.

* FIGURE 22. Output crosstalk (low-level and high-level) test circuit for table XVI.

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NOTES:
1. Input pulse parameters:
   $t_{THL} = t_{TLH}$ = 3.5 ±1.5 ns (verified at each input pin).
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. The lead length of electrical conductors between terminator resistor R1 and input pins; probes Y01-1, Y01-2 to input and output pins; load circuits to output pins; and ground leads of probes Y01-1, Y01-2 to test fixture and module ground pins shall be less than 1 inch (25.4 mm).

* FIGURE 23. Output crosstalk (low-level and high-level) test circuit waveforms for table XVI.

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TABLE XVII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
U	Power supply transient amplitude	0	400	mV	24,25

TABLE XVIII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V	Power supply filter capacitance	Open	1	1.6	3.0	µF	26

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* TABLE XIX. Input capacitance test.

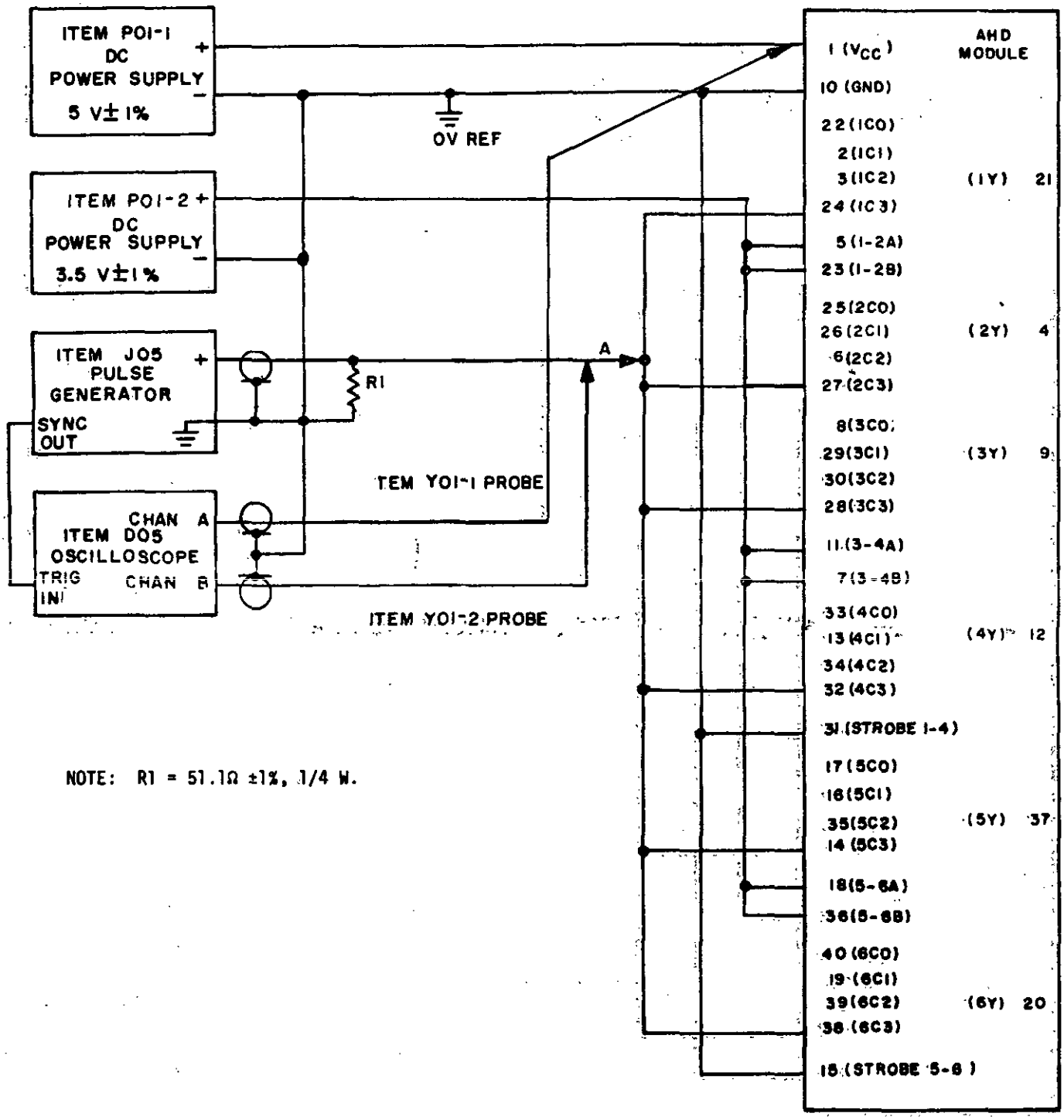
Test condition	Parameter		Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
W1	1C0	Input capacitance	22	Open	2	15	pF	26
W2	1C1		2					
W3	1C2		3					
W4	1C3		24					
W5	2C0		25					
W6	2C1		26					
W7	2C2		6					
W8	2C3		27					
W9	3C0		8					
W10	3C1		29					
W11	3C2		30					
W12	3C3		28					
W13	4C0		33					
W14	4C1		13					
W15	4C2		34					
W16	4C3		32					
W17	5C0		17					
W18	5C1		16					
W19	5C2		35					

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* TABLE XIX. Input capacitance test - continued.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
W20	5C3	Input capacitance	14	Open	2	15	pF	26
W21	6C0		40					
W22	6C1		19					
W23	6C2		39					
W24	6C3		38					
W25	1-2A		5					
W26	1-2B		23					
W27	3-4A		11					
W28	3-4B		7					
W29	5-6A		18					
W30	5-6B		36					
W31	STROBE 1-4		31			45		
W32	STROBE 5-6		15			30		

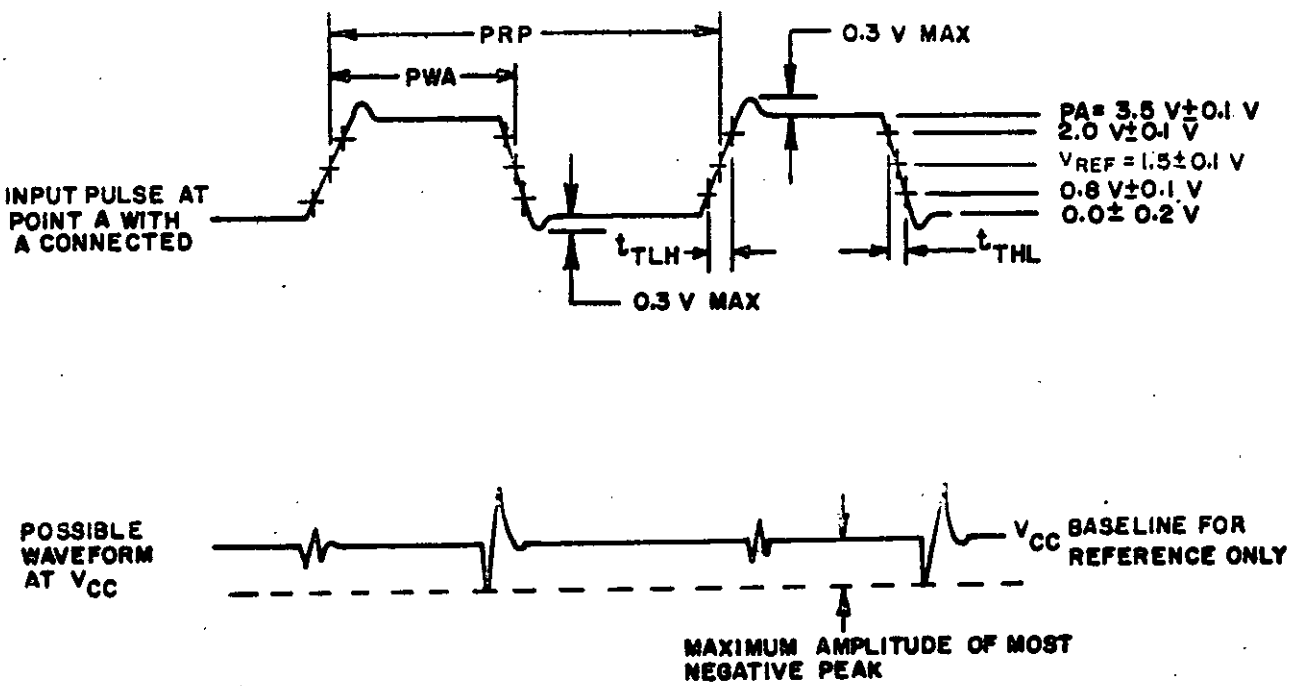
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NOTE: R1 = 51.1Ω ±1%, 1/4 W.

* FIGURE 24. Power supply transient amplitude test circuit for table XVII.

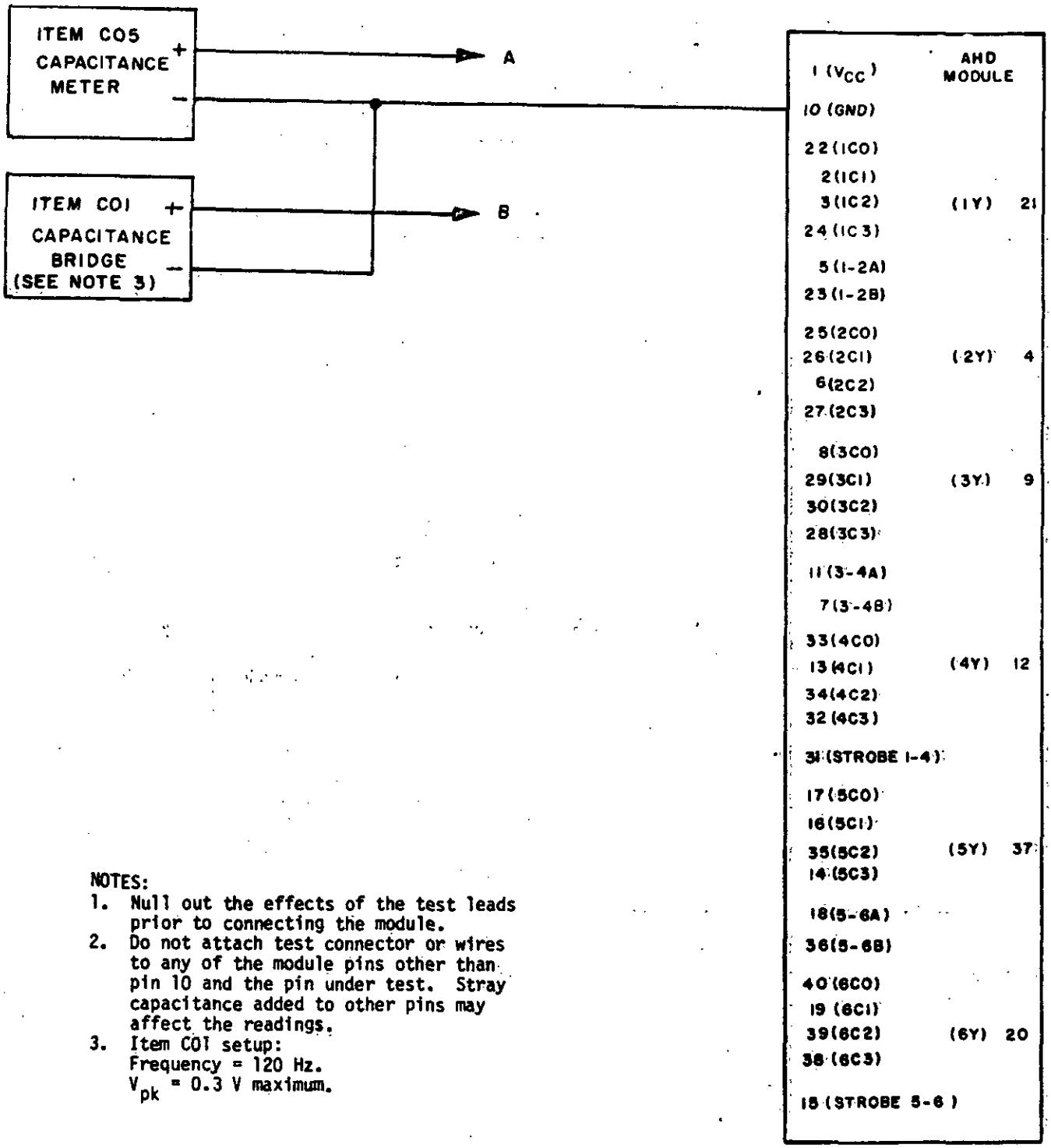
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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
PWA = 500 ±50 ns.
PRP = 1,000 ±100 ns.

* FIGURE 25. Power supply transient amplitude test circuit waveforms for table XVII.

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

* FIGURE 26. Power supply filter capacitance and input capacitance test circuit for tables XVIII and XIX.

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